COMP1609 Network and Internet Technology and Design

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Task 1:

Introduction

The purpose of this task is to analyse the potential risks arising from technologies such as DevOps, Cloud Computing and Data Security. The task will be carried out by two members; The risk assessor will evaluate the potential threats and the upgrade engineer will provide a benefit point of view.

Since early integration of security into the development process can potentially reduce the likelihood of security flaws, improve cooperation between the security and development teams, facilitate quick reactions to security events, boost automation, and guarantee regulatory compliance, security in general becomes an essential part of DevOps and Agile.

Organizations are attempting to deploy software at a faster speed, which increases the risk of security flaws and breaches, according to research on DevOps. However, there are problems with authorization and authentication since security measures for authentication cookies have been deployed incorrectly, incompletely, or not at all. Therefore, weak authentication of cookies may lead to vulnerabilities in DevOps, just like the CVE-302, which allows bypassing authentication by assuming immutable data, where intruders can gain access to features of an authenticated session.

-Mitre.org. (2021). CWE - CWE-302: Authentication Bypass by Assumed-Immutable Data (4.18). [online] Available at: https://cwe.mitre.org/data/definitions/302.html.

Risk assessment based on the example

- Keep a track of two bits of taint for each data within the application: a "credential" taint bit, which signifies if the data item corresponds to a recognized valid password or credential, and a "user input" taint bit, which denotes whether the data item was provided by the user as part of the HTTP request. User input encompasses all values supplied by the untrusted client, including HTTP request headers, cookies, POST bodies, and URL parameters.
  (Wei, Z., Shi, K. and Jia, D. (2011). A conceptual model for preventing web bypass vulnerabilities.)
  Implement proper protection for immutable data (e.g., environment variables, hidden form fields, etc.).

Cloud Computing

Cloud security is an extensive cybersecurity domain encompassing the tools, technologies, policies, and controls necessary to safeguard data, applications, and infrastructure within cloud computing environments. Threats such as data breaches, malware, and unauthorized access remain prevalent, as illustrated by the widespread exploitation of common misconfigurations in public S3 buckets, highlighted by CVE-2024-50764, which indicates that the use of Jenkins Scriptler Plugin version 342.v6a_89fd40f466 and earlier versions allowed attackers to manipulate files indiscriminately.

This allows attackers to send an HTTP request to a user-specified URL and interpret the response as XML.
Risk assessment based on the example

- The Analysis Model API Plugin requires an update to version 11.13.0.
- The Nexus Platform Plugin requires an update to version 3.18.1-01.
  The Scriptler Plugin requires an upgrade to version 344.v5a_ddb_5f9e685.
- *(Jenkins Security Advisory 2023-12-13. 2023)*.

Upgrade Engineer:
Using the above example DevSecOps could of help mitigated this.
The vulnerability is described as "Authentication bypass by assumed-immutable data." Which could have been avoided by using the tools in the DevOpsSec toolchain.

For instance, Threat modelling can Identify, Analyse, Mitigate and Test. *(GeeksforGeeks 2018)*
We can also use Static Application Testing (SAST) or Dynamic Application Testing (DAST). *(Secopsolution.com. (2025))*
Code review using a security expert could have also been able to see this issue before it entered the Pipeline.
One other method that could be used is Penetration testing by running real-world attacks, this could have been done by external testers or in-house using Red Team methodology. *(SecurityScorecard. 2025).*

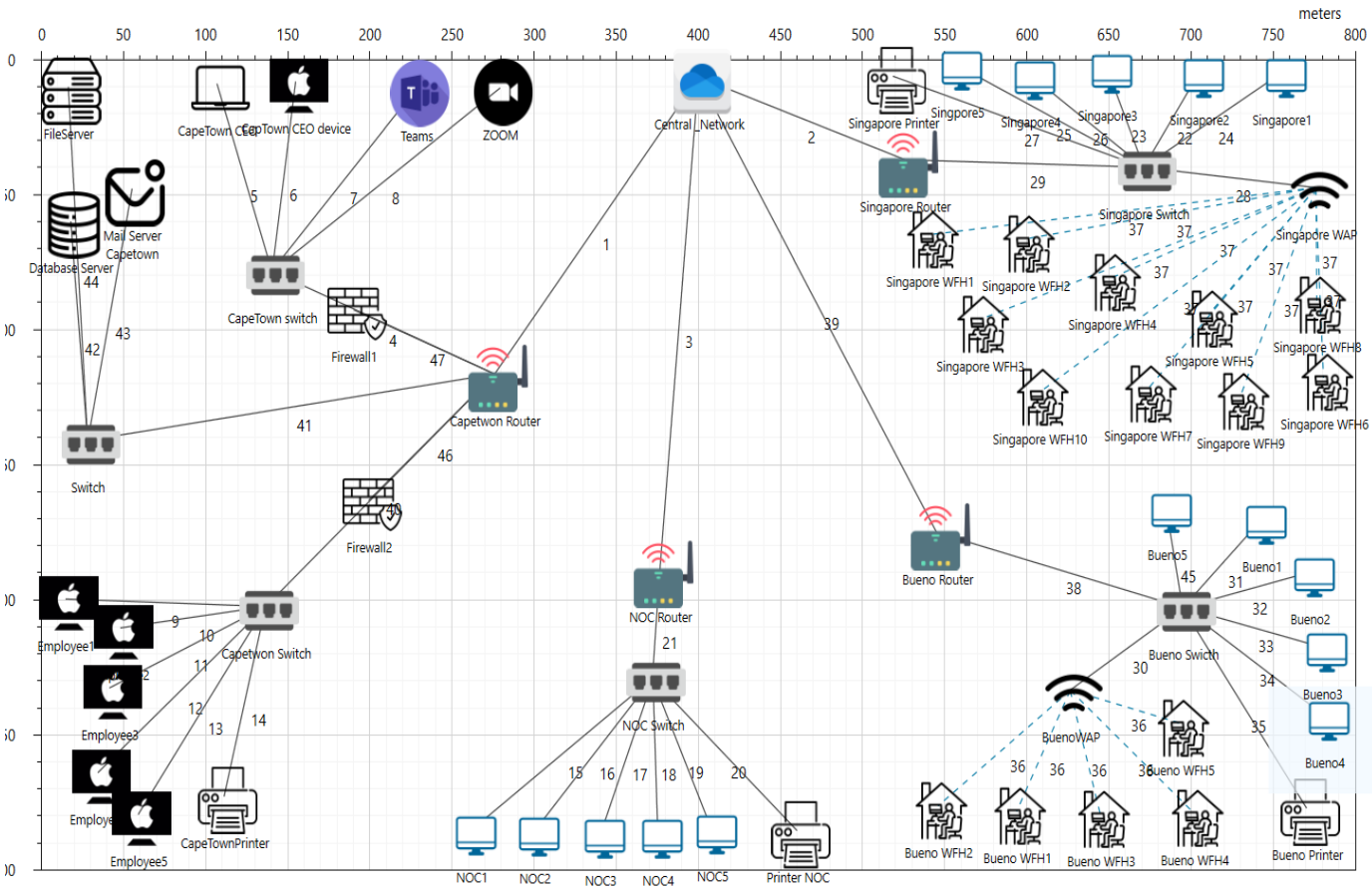
This vulnerability happened with the Azure Cloud environment and all the above could have stopped this issue before the services were built within Azures.

Task 2

• The task involves designing and modelling a global network architecture for four offices: the main office in Cape Town, South Africa; regional offices in Buenos Aires, Argentina, and Singapore; and a Network Operations Centre (NOC).
• The network will support various applications, including voice, video, email, FTP, and databases, aiming for a high-level design.
• The topology will feature distinct segmentation focusing on connection, performance, and security.

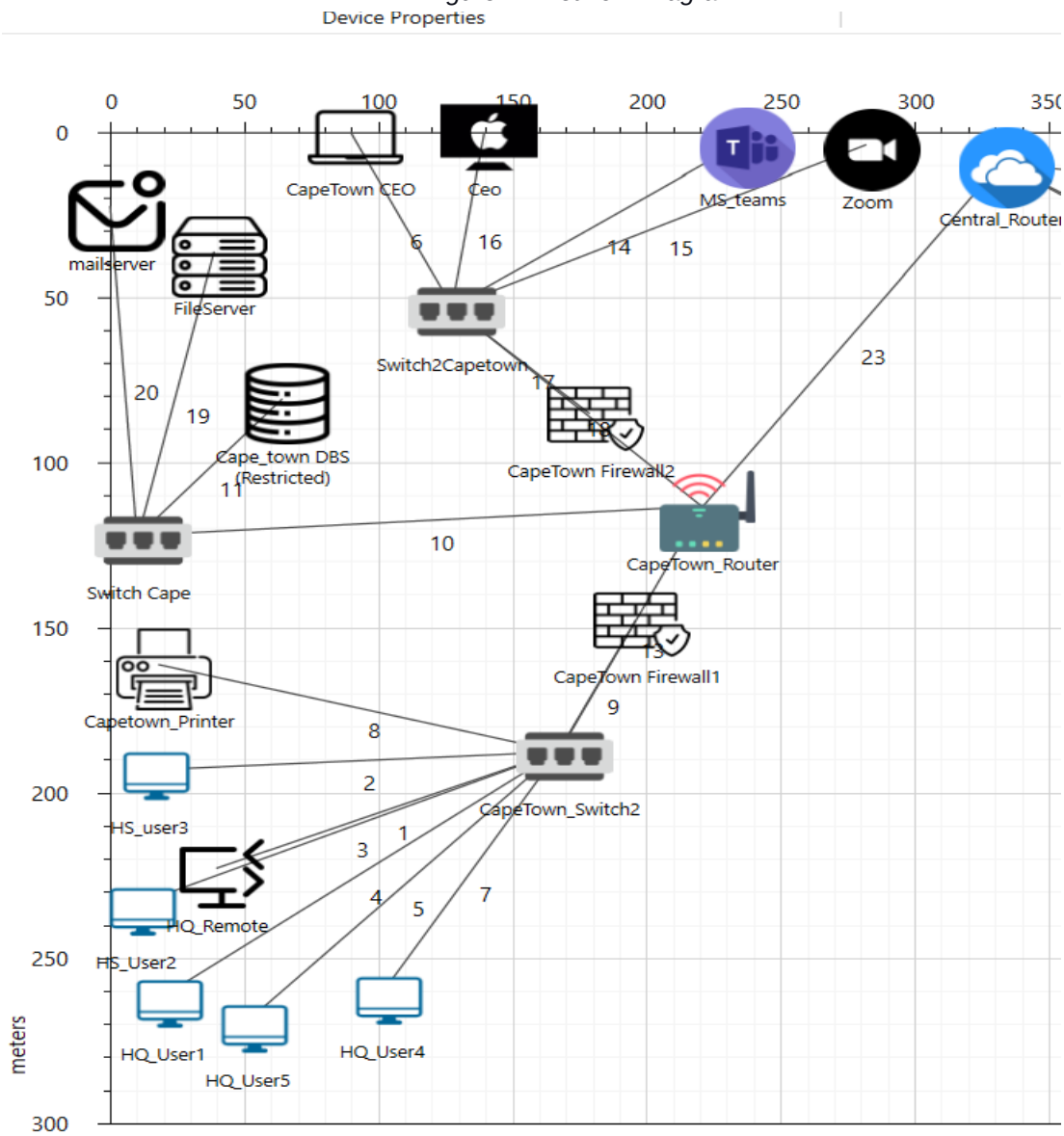
Topology Overview

Figure 2.1 shows the overall topology



CAPE TOWN NETWORK

Figure2.2 Network Diagram



• The Cape Town network is organized similarly to a conventional corporate headquarters.
• It features a robust network security mechanism, implemented through effective network segmentation.
• The architecture comprises three distinct zones: CEO Zone, Server Zone, and Employee Zone.
• Two firewall nodes serve as logical security barriers to filter traffic between these zones.
• Switches are used to represent the firewall, despite the absence of physical firewall hardware in NetSim.

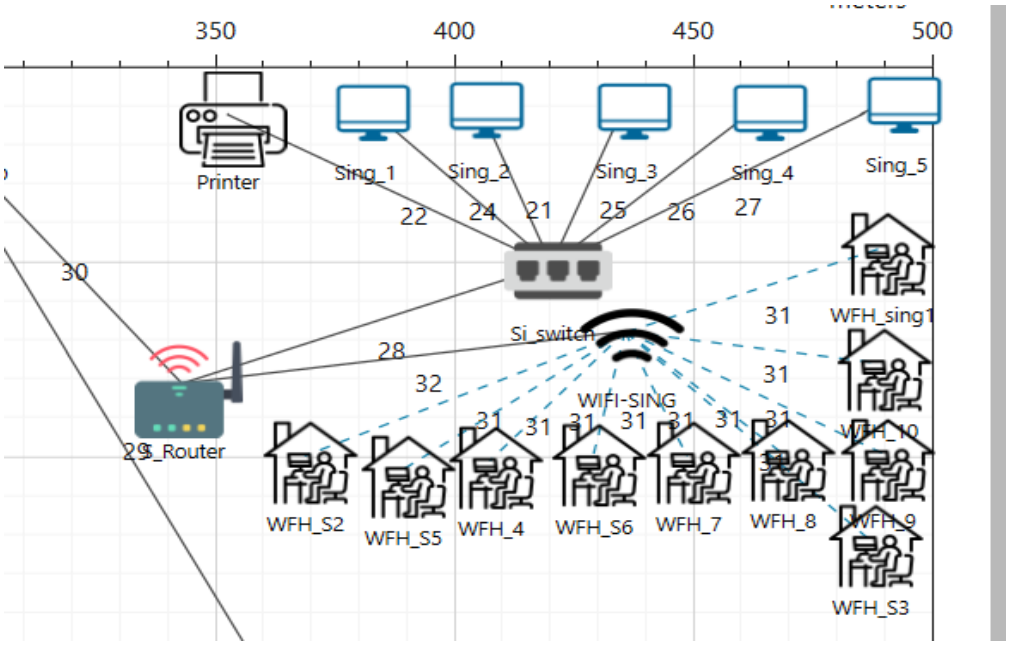
To filter traffic between zones, two firewall nodes are positioned as logical security barriers. Switches are utilised symbolically to more fully reflect the firewall, even though NetSim does not include physical firewall hardware.
This is an example of a VRF-based segmentation that ensures symmetric traffic flow within the zones by filtering traffic from external branches and between internal zones (Sebakor, Mahamah 9 May 2023).

• Various applications, including FTP, mail, database, and MS Teams for voice applications, operate between server nodes and end hosts across the zones.
• LAN lines were adjusted for a propagation delay of 5 ms to simulate high-speed communication.

Singapore Network

Diagram

(Figure 2.3 representing Singapore Network)

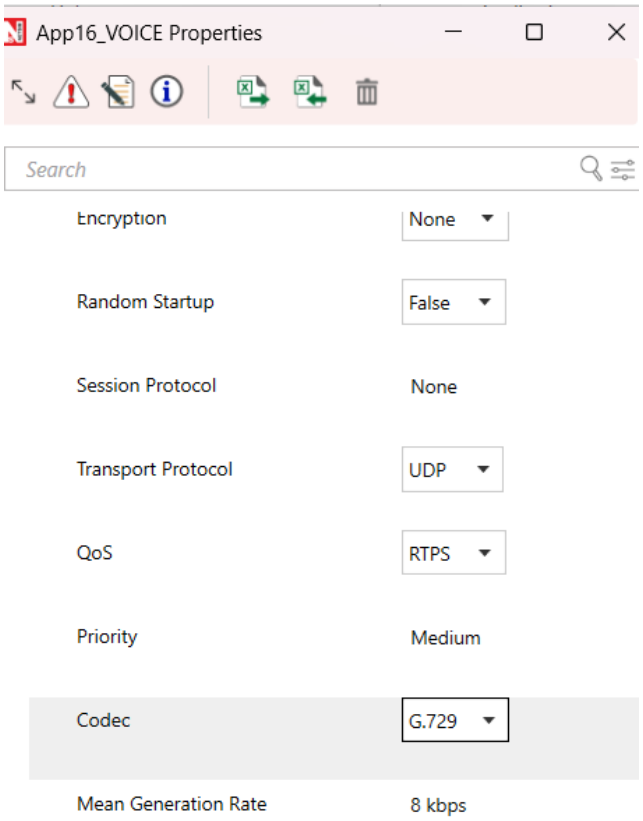


The Singapore network is categorised as a SOHO-type network, resembling a small office or work-from-home setup.
• It features a flexible and cost-effective design, integrating both wired and wireless users.
• The network comprises a simpler LAN architecture with a primary switch, a wireless access point (WAP), a printer, and various wired and wireless nodes.
• It simulates a scenario where 5 wired users can access a local printer, and 5 wireless users can participate in Zoom voice calls for meetings hosted by CapeTown Headquarters.

Application

- Voice applications: encoded in codec =G.729 with inter-arrival time= 20ms for clear audio experience
- Mail Applications: set to TCP with exponential distribution of packet size 200B/8000B

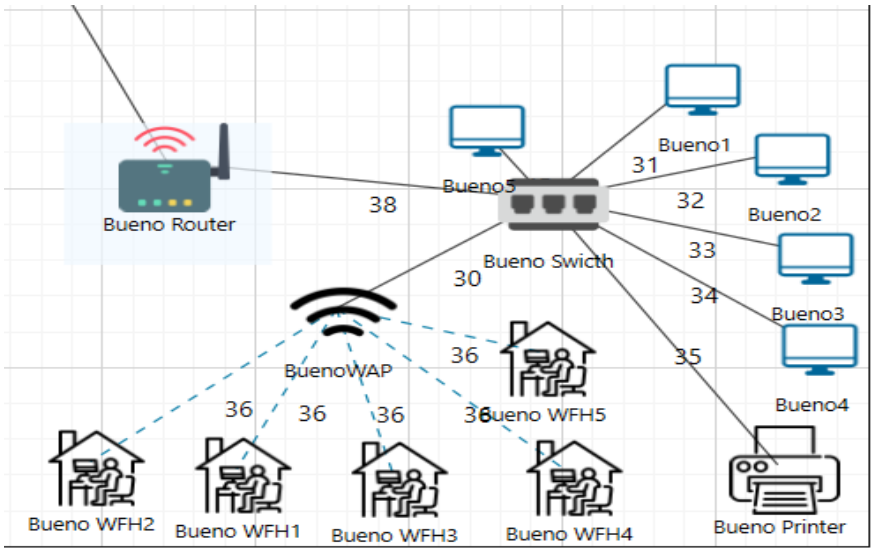
Figure 2.4 shows changes to the voice application



LAN links are set to 5ms Uptime/Downtime
WAN link set to 20ms Uptime/Downtime

Bueno Network
Diagram

(Figure 2.5) Topology represents the Bueno network



- The Bueno network is set up similarly to the Singapore scenario, consisting of a users connecte to BUE_SW2 and BUE_WAP.
- The Wireless nodes connected to the WAP have access to MS Teams via video calls and voice calls by attending meetings hosted by CapeTown Headquarters.
- Applications
- Voice application: configured at Qos ERTGS encoded in default G.7211 for effective communication.
- Mail Applications: set to TCP with exponential distribution of packet size 200B/8000B

Links

- LAN links: propagation delay 2ms/2ms
- WAN links: propagation delay 10 ms/ms

Network Operations Centre.
A typical LAN-based small office network representing wired nodes, a printer switch and a router
Users' access to local printers, Microsoft Teams via voice and video

Applications

- Video call: codec G.729 Qos: ERTGS
- Voice call: Default

Links

- LAN links: propagation delay 5ms/5ms
- WAN links: propagation delay 10 ms/ms

(Figure 2.6 topology represents the NOC network

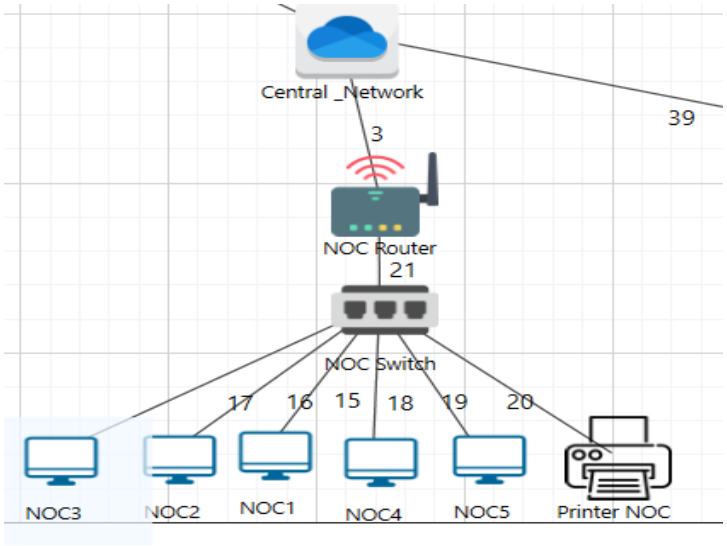
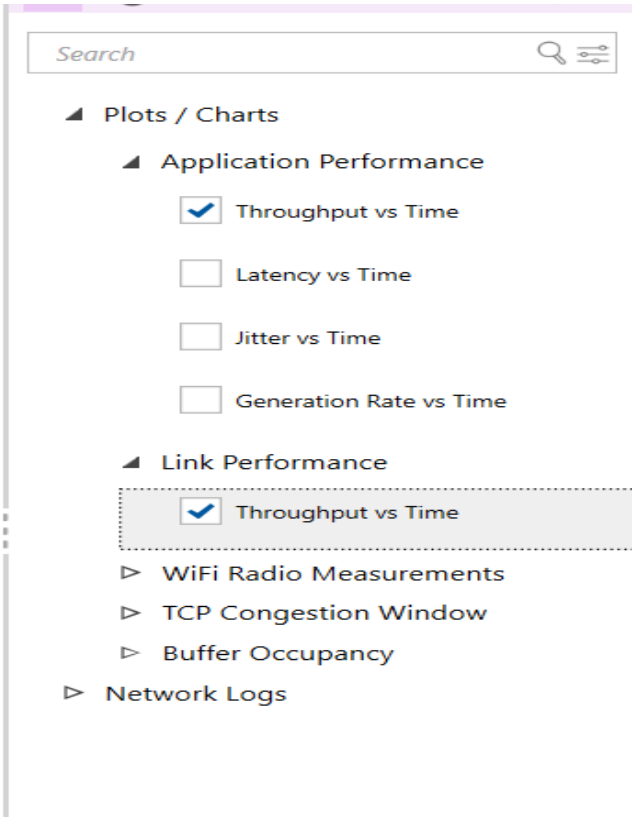


Figure 2.7 shows configurations made



TASK 3

IP Addresses and Subnets of devices in Cape Town

In this section, we will provide a tabular representation of LAN/WAN/MAN with IP addresses, Subnet, network addresses, broadcast address and IP assignment to devices

1. CAPE TOWN (HEADQUARTERS)

Location and class	IP address	Network address	Broadcast Address	Subnet mask	Usable address range
CapeTown LAN (class C)	192.168.20.0/24	192.168.20.0	192.168.20.255	255.255.255.0	192.168.20.1-192.168.20.254
CapeTown WAN(Connecting to Central Network) (Class-B)	172.16.1.0/30	172.16.1.0	172.16.13	255.255.255.252	172.16.1.1-172.16.1.2

IP Assignment to devices connected to CapeTown LAN

Device and branch	Ip Address
Employee1	192.168.20.10
Employee2	192.168.20.11
Employee3	192.168.20.12
Employee4	192.168.20.13
CapeTown Printer	192.168.20.14
Mail Server	192.168.20.16
Databse Server	192.168.20.18
File Server	192.168.20.21
CapeTown CEO Device	192.168.20.19
Capetown CEO	192.168.20.15
Firewall1	192.168.20.22
Firewall2	192.168.20.20

`

2. Singapore LAN

Location and class	IP address	Network address	Broadcast Address	Subnet mask	Usable address range
Singapore LAN (class C)	192.168.40.0/24	162.168.40.0	192.168.40.255	255.255.255.0	192.168.40.1-192.168.40.254
Singapore WAN (Class-B)	172.16.1.8/30	172.16.1.8	172.16.1.11	255.255.255.252	172.16.1.9-172.168.16.1.10

IP Assignment to devices connected to Singapore LAN

Device and branch	Ip Address
Singapore1	192.168.40.1
Singapore2	192.168.40.2
Singapore3	192.168.40.3
Singapore4	192.168.40.4
Singapore5	192.168.40.5
SingapreWFH1	192.168.40.10
SingaporeWFH2	192.168.40.11
SingaporeWFH3	192.168.40.12
SingaporeWFH4	192.168.40.13
SingaporeWFH5	192.168.40.14
SingaporeWFH6	192.168.40.15
SingaporeWFH7	192.168.40.16
Singapore WFH8	192.168.40.17
Singapore WFH9	192.168.40.18
Singapore WFH10	192.168.40.19

3. Buenos Network

Location and class	IP address	Network address	Broadcast Address	Subnet mask	Usable address range
Buenos LAN (class C)	192.168.30.0/24	192.168.300	192.168.30.255	255.255.255.0	192.168.30.1-192.168.30.254
Buenos WAN (Class-B)	172.16.1.4/30	172.16.1.4	172.16.1.7	255.255.255.252	172.16.1.5-172-172.16.1.6

IP Assignment to devices connected to Buenos LAN

Device and branch	Ip Address
Bueno5	192.168.30.3
Bueno2	192.168.30.4
Bueno3	192.168.30.6
Bueno4	192.168.30.7
Bueno printer	192.168.30.2
BuenoWFH1	192.168.30.8
BuenoWFH2	192.168.30.15
BuenoWFH3	192.168.30.13
BuenoWFH4	192.168.30.11
BuenoWFH5	192.168.30.14

4. NOC Network

Location and class	IP address	Network address	Broadcast Address	Subnet mask	Usable address range
NOC LAN (class B)	172.16.10.0/29	172.16.10.0	172.16.10.7	255.255.255 .248	172.16.10.1-172.16.10.6
NOC WAN (Class-B)	172.16.0.0/30	172.16.0.0	176.16.0.3	255.255.255 .252	172.16.0.1-172.16.0.2

IP assignment to devices connected to the NOC network

Device and branch	Ip Address
NOC1	176.16.10.1
NOC2	176.16.10.2
NOC3	176.16.10.3
NOC4	176.16.10.4
NOC5	176.16.10.5
Printer NOC	176.16.10.6

Central Network

Location and class	IP address	Network address	Broadcast Address	Subnet mask	Usable address range

NOC WAN (class B)	172.16.0.0/30	172.16.0.0	172.16.0.3	255.255.255.252	176.16.0.1-176.16.0.2
Buenos WAN	172.16.1.4/30	172.16.1.4	172.16.1.7	255.255.255.252	172.16.1.5-172.16.1.6
CapeTown WAN	172.16.1.0/30	172.16.1.0	172.16.1.3	255.255.255.252	172.16.1.1-172.16.1.2
Singapore WAN	172.16.1.8/30	172.16.1.8	172.16.1.11	255.255.255.252	172.16.1.9-172.16.1.10
MS Teams/Zoom	NAT	NAT	NAT	NAT	NAT

TASK 4

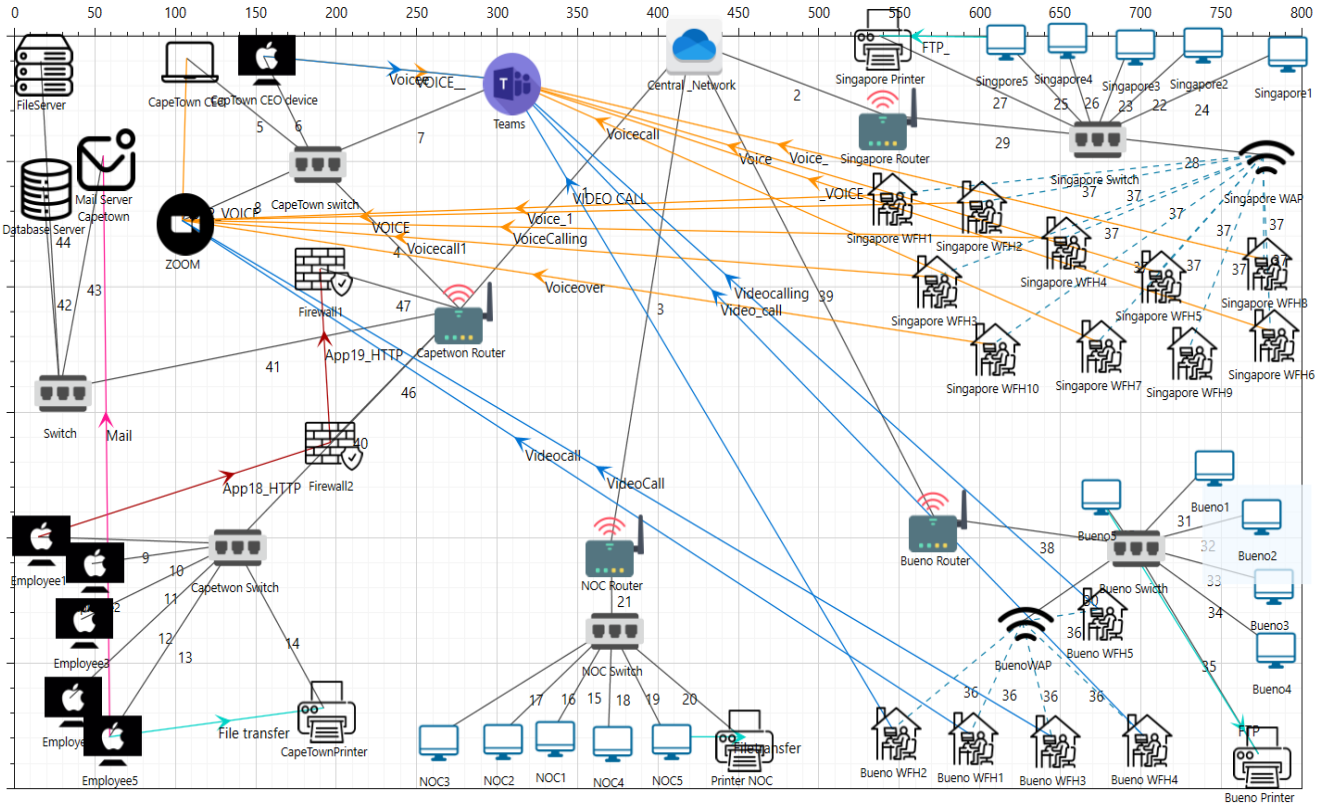
In this task, we look at how apps work on a multi-branch enterprise network, such as the Cape Town Headquarters Network, Singapore, Buenos, and NOC. apps like FTP, Email, Database, and Voice/video run on both LAN and WAN connections. We will study critical metrics like as jitter, latency, packet loss, and throughput. We established objectives for each scenario and subsequently conducted simulations to obtain Netsim data on metrics such as Throughput versus Time, Latency versus Time, Jitter/Packet Loss, and Link Utilisation. Subsequently, we examine the outcomes to determine if they align with our expectations

The task will aim to assess the behaviour of applications based on a list of scenarios. These analyses include

- Packet-level behaviour
- Delays
- WAN vs LAN performance variations

Application Flow Diagram

Figure 4.1 diagram shows Application traffic across the branches



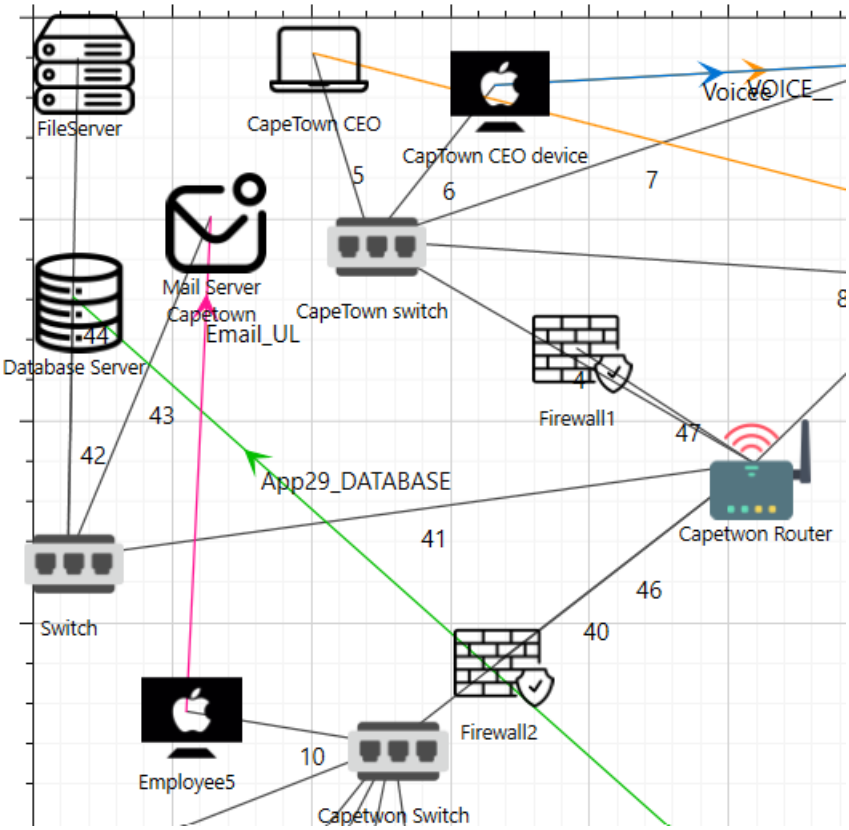
Scenarios Considered

1. Inter-branch email traffic within the Cape Town network compared to intra-branch communication to the NOC network to illustrate LAN versus WAN. Behaviour in relation to the application
2. Comparison of LAN and WAN Links in Relation to Overall Topology

Scenario 1:

1.1 Demonstrating LAN behaviour in the Cape Town Network by sending Inter-branch Email traffic VS Intra-Branch

Figure 4.2 shows the Mail application traffic

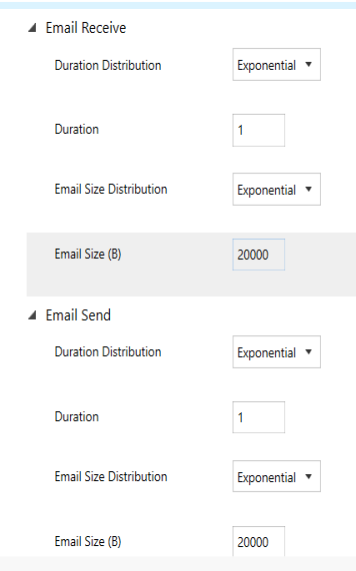


As per **Figure 4.2**, we see that a mail application with the application name **Email_UL** is being sent from **employee5** to the **mail**

Expectations for the traffic

App_name	Application type	Traffic Flow	Link Type	Changes	Justification
Email_Ul	Mail	Employee5 to mail server (Capetown)	LAN	Exponential Packet Size 2000/ 8000 (bytes)	To ensure reliable communication, we set TCP length standards, as emails can show burstiness due to fluctuating text sizes between 2 mb and 8 mb depending on user load.
EMAIL!	Mail	Employee1 to NOC1 (CapeTow n-NOC)	WAN	Exponential Packet size 10000/20000 Bytes	The email size distribution follows an exponential model, encompassing sizes between 10MB and 20MB, due to WAN transfers including extra headers, authentication overhead, and larger attachments.

Figure 4.3 shows configuration made



Based on the following expectations, we would run the simulation for 20s and plot application throughput vs time

Figure 4.4 below shows the application throughput vs time result

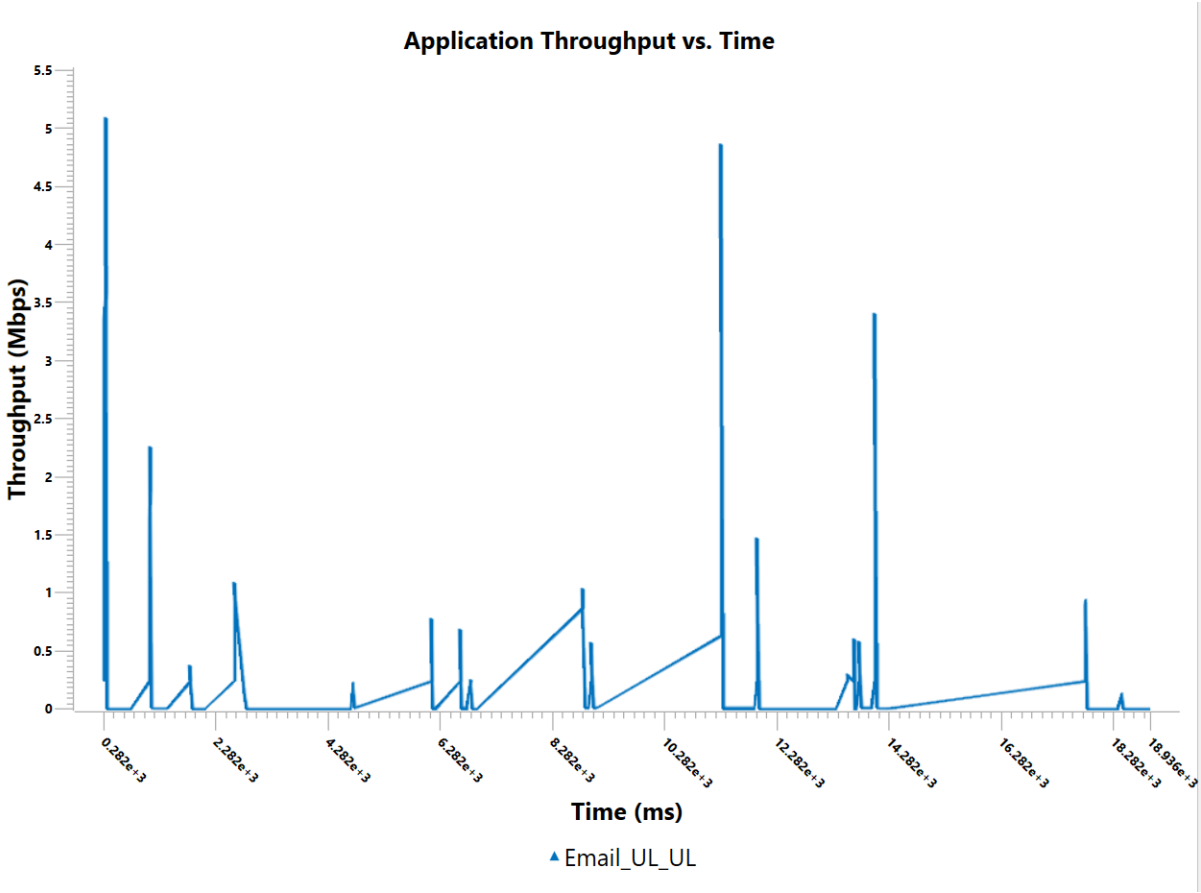


Figure 4.5 below shows the Link throughput vs time result

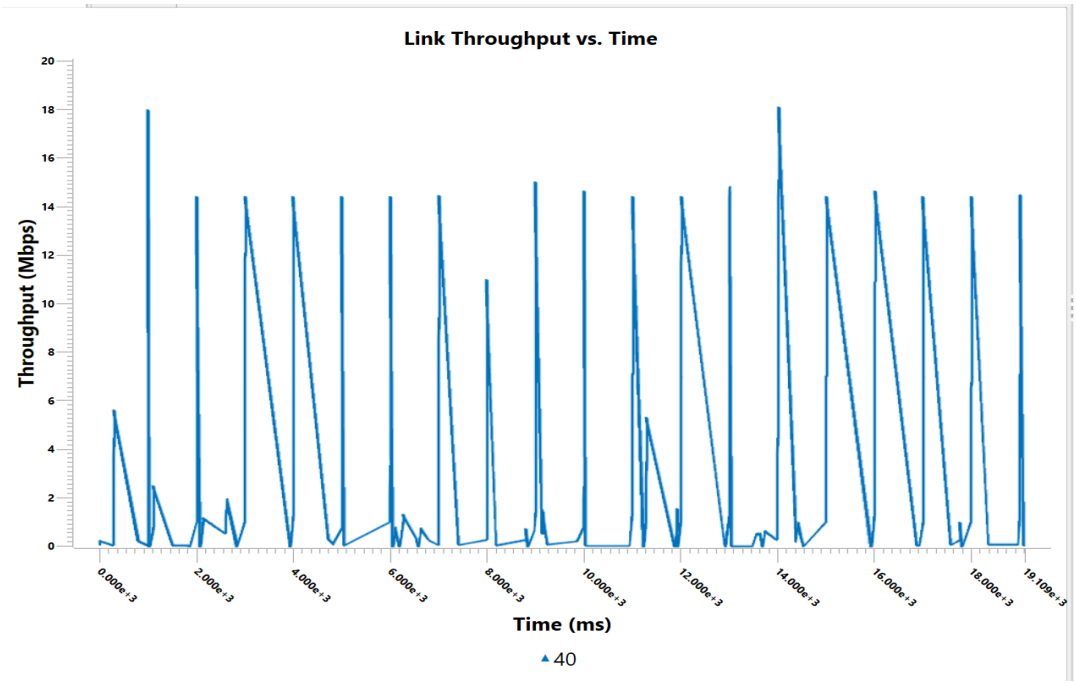


Figure 4.5 below shows the application Latency vs time result

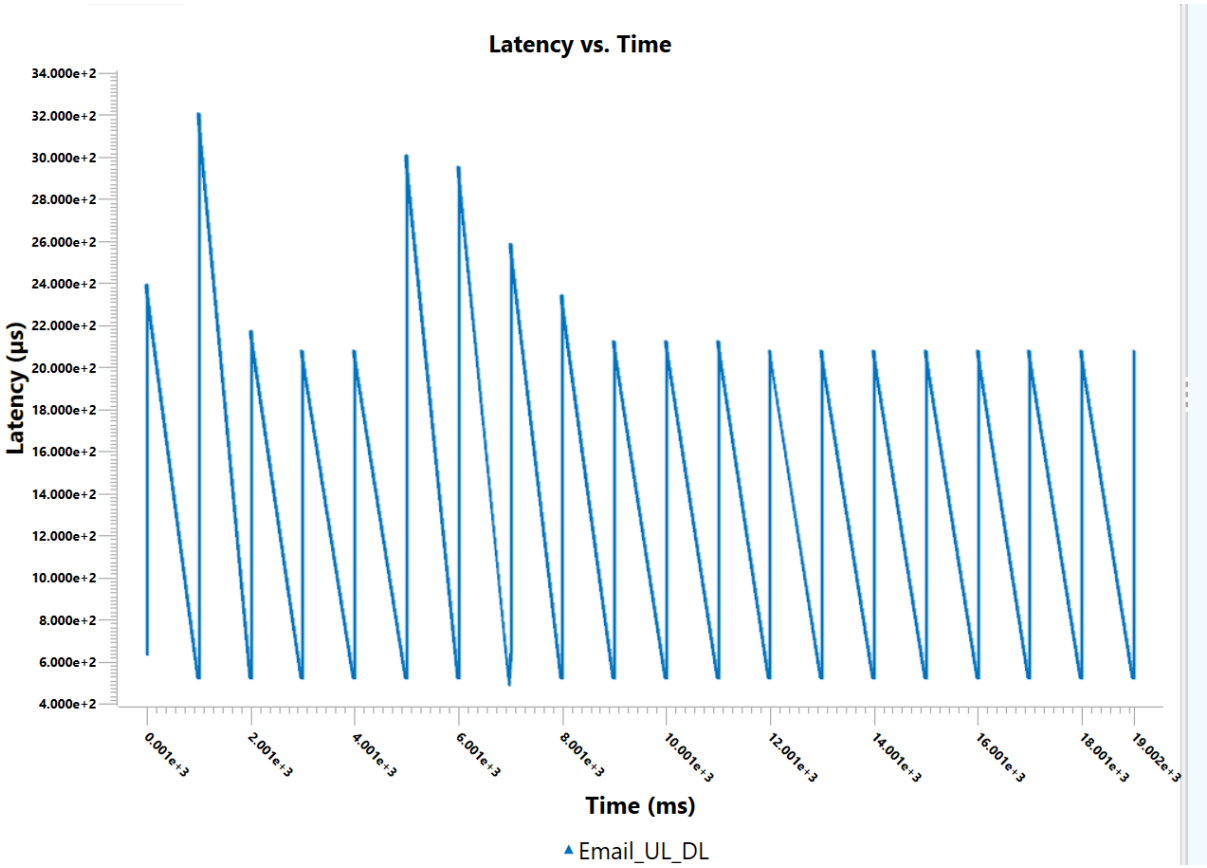
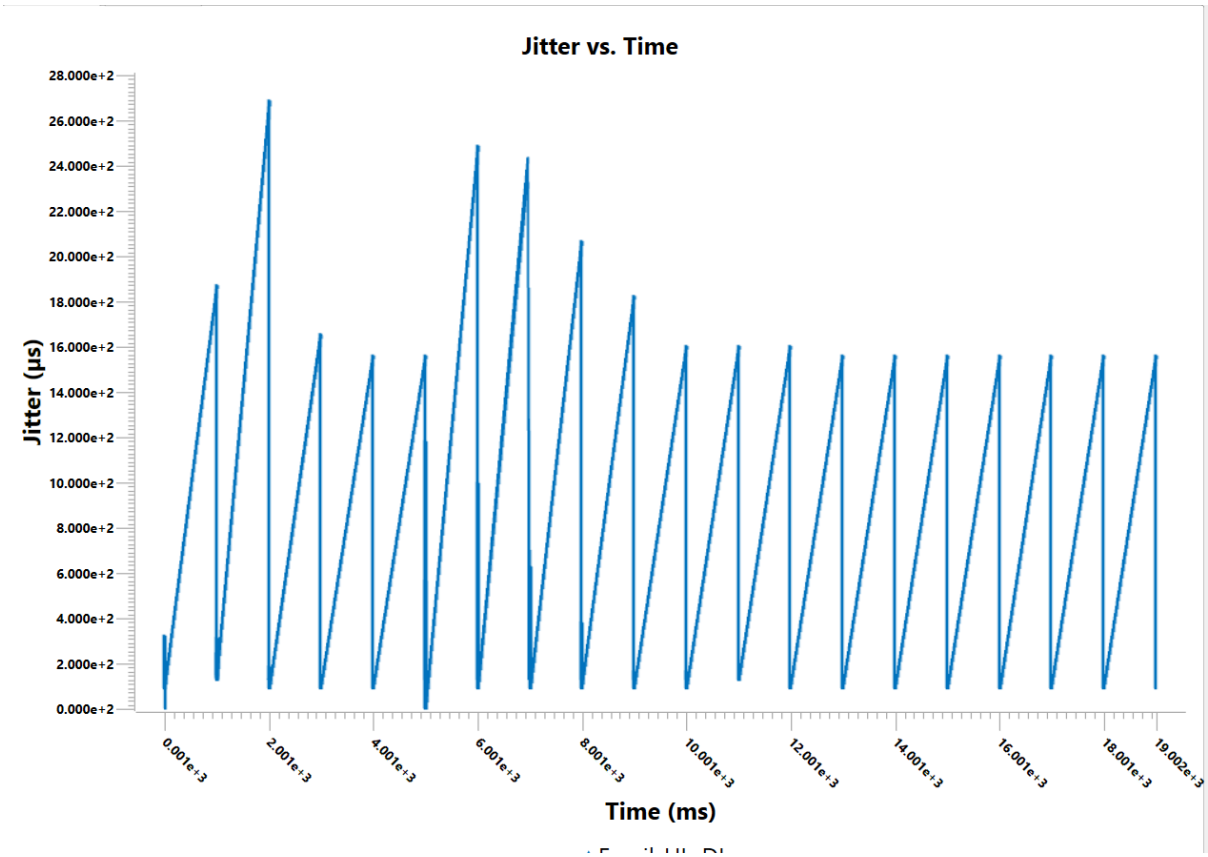


Figure 4.4 below shows the application jitter vs time result



Application Metrics

	Application	Packets generated	Packets received	Throughput(Mbps)	Delay microsecond	Jitter
18	Email_UL	280	280	0.16	2165.816	410.755
18	Email_UL	133	133	0.076	1042.365	235.8861

Link Metrics

Link	Bytes transmitted(B)	Payload transmitted(B)	Overhead transmitted(B)
40	2390240	2163900	226340

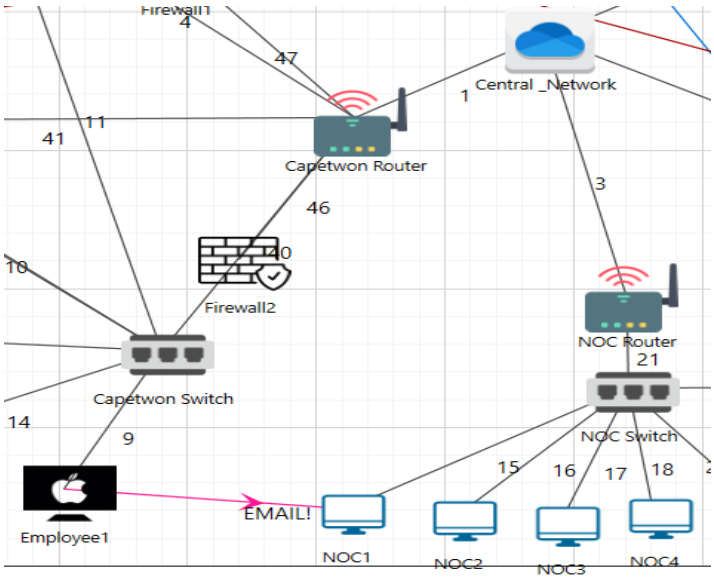
Although the **Email_UL** application was configured in exponential distribution, the application **Throughput vs Time plot** shows repeated drops between bursts. This could be due to per-flow TCP dynamics or processing/queing delay arising in **Link40** from **Firgue4.5**, which displays continuous, consistent drops

Typically, LAN communications exhibit great throughput and minimal delays; however, this circumstance presents a stark contrast.

VS

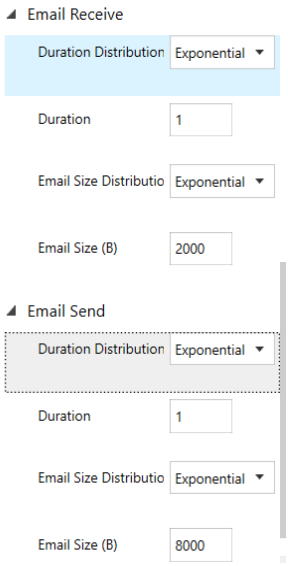
1.2 Sending intra-branch traffic of mail from CapeTown user to the NOC user

Figure 4.5 inter-branch traffic sceanrio



As per **Figure 4.5**, we see that a mail application with the application name **EMAIL!** is being sent from **employee1** to the **NOC1**

Figure 4.6 below shows configurations made



Based on the following expectations, we would run the simulation for 20s and plot application throughput vs time

Figure 4.7 shows the application throughput vs time result

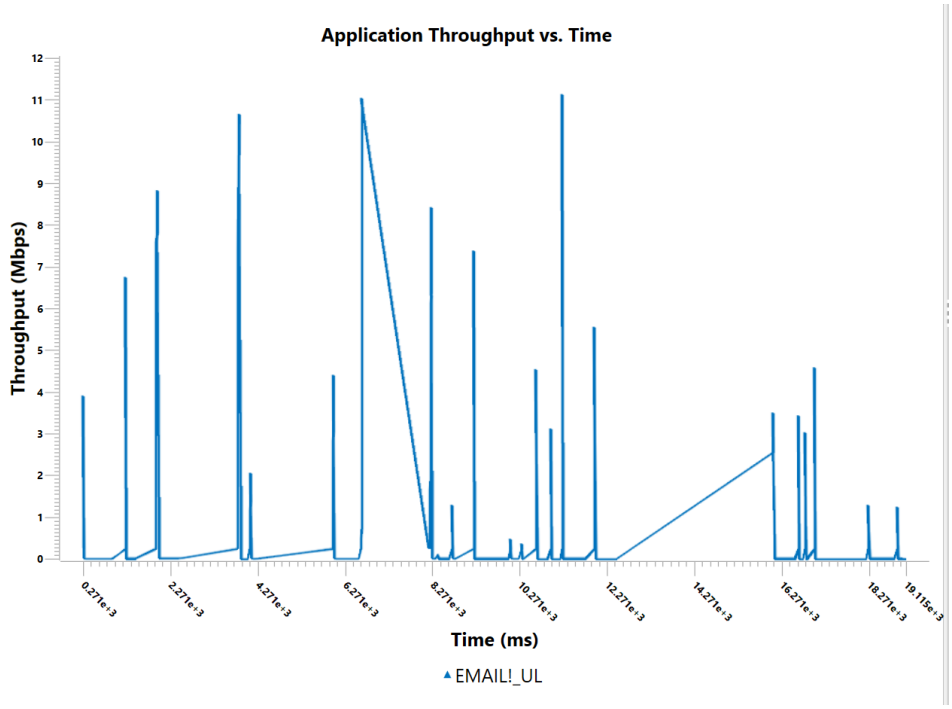


Figure 4.8 below shows the application throughput vs time result

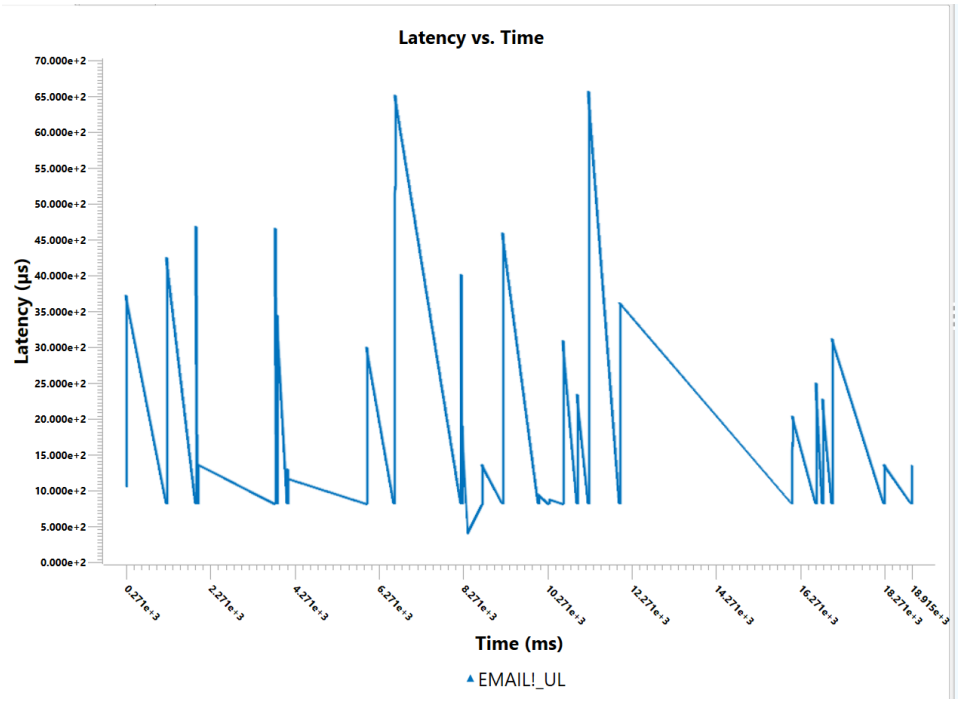
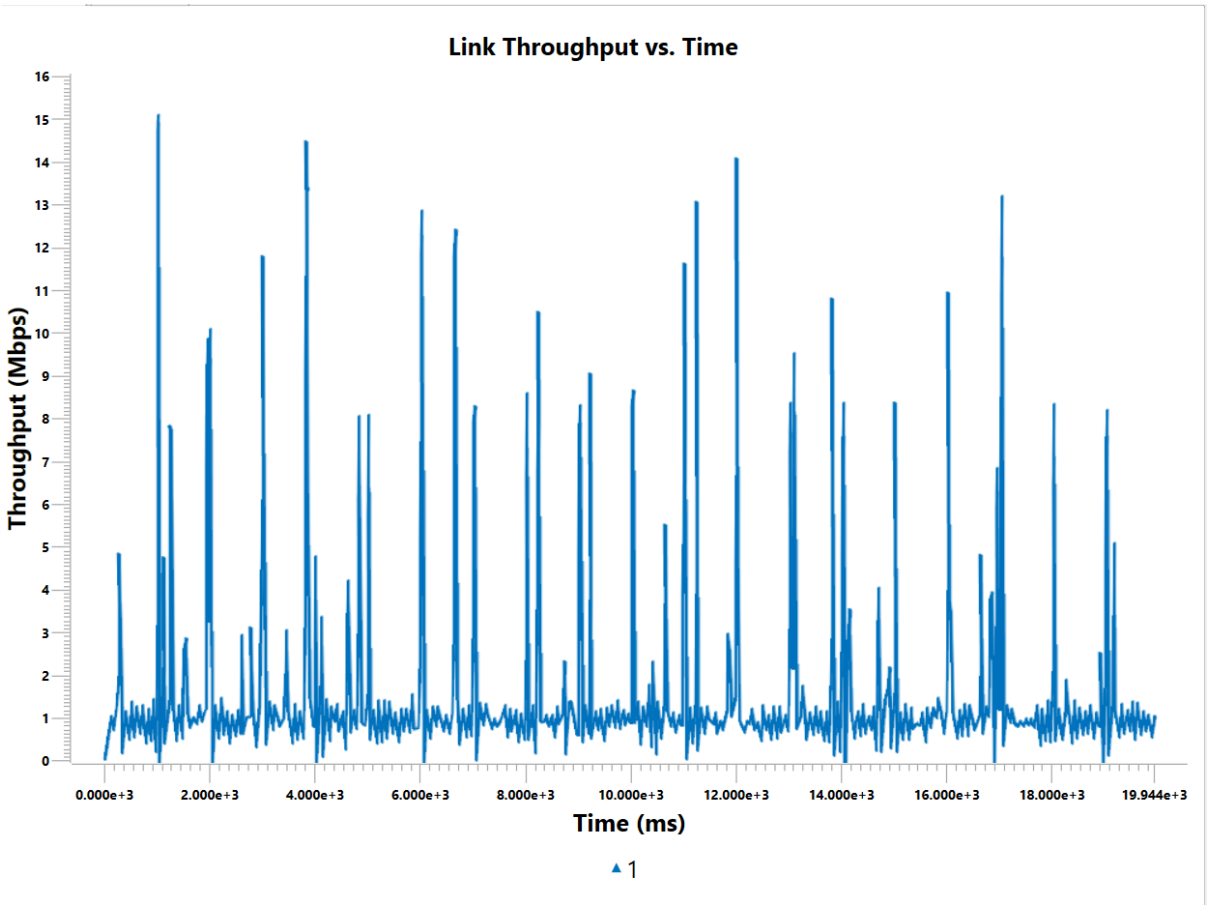


Figure 4.8 below shows the link throughput vs time result



Application Metrics

	Application	Packets generated	Packets received	Throughput(Mbps)	Delay microsecond	Jitter
19	Email_UL	471689	471689	0.188676	95830.40	9235.77
19	Email_UL	674336	674336	0.269734	2463.37	232.73

Link Metrics

Link	Bytes transmitted(B)	Payload transmitted(B)	Overhead transmitted(B)
1	4420390	4017906	402484

The results indicate that **EMAIL!** The application sent to **NOC1** exhibited strong overall throughput and substantial byte volume; however, it experienced significant collisions, potentially attributable to extended WAN distances and associated propagation delays resulting in retransmissions.

Key takeaways for this scenario
LAN EMAIL vs WAN EMAIL!

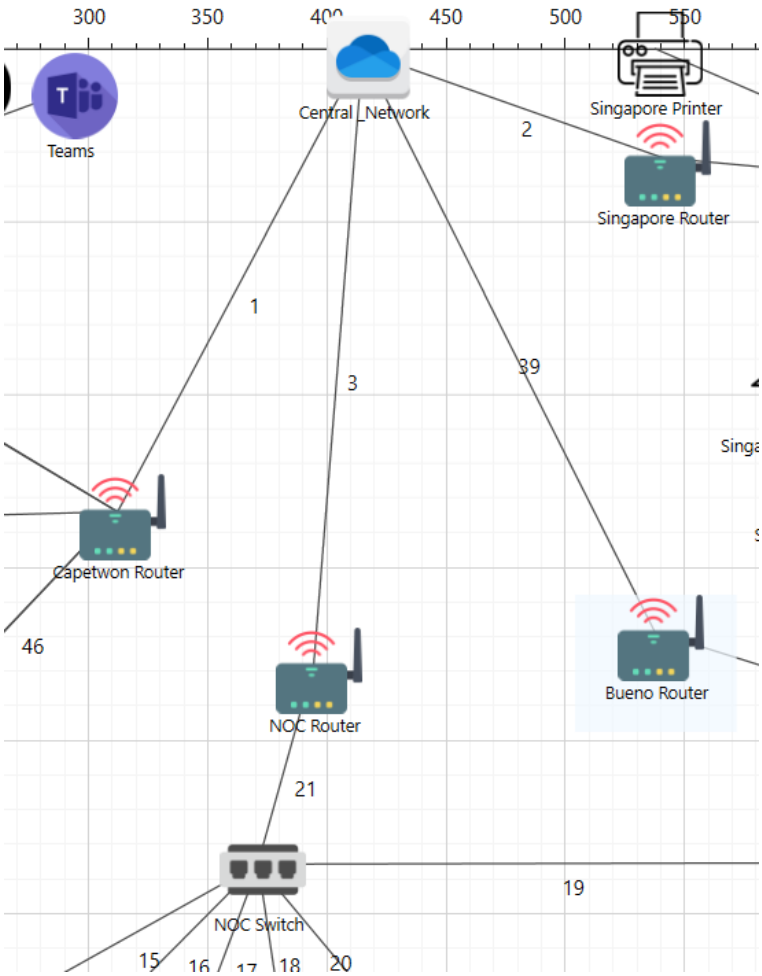
Comparison	LAN **EMAIl**	WAN **EMAIL!**	Notes
Total Volume Traffic	• Bytes Transmitted 2390240(B) • Payload: 2163900(B) • Overhead 226340	Bytes transmitted 4420390(B) Payload transmission 4017906(B) Overfead transmission (B) 402484	WAN Transimtted 17 times more than the LAN while lan traffic was light and short and more acks
Overhead Comparison	Only 226KB overhead with a few headers	6.5mb of overload	Overhead increases drastically moving from a LAN to WAN
Throughput Behaviour	Throughput fluctuation at 1-5 Mbps-bursty with an inconsistent pattern	Throughput fluctuation yet consistent	WAN email showed high throughput, consistency, and stability, while LAN email had inconsistent throughput.

Efficiency ratio (Payload /Total bytes)	Efficiency= 2163900(B)/2390240(B)=90.5%	Efficiency = 4017906(B)/4420390(B)=90.8%	Surprisingly, WAN demonstrated greater efficiency due to proportional overhead and fewer collisions.

1. Scenario

Scenario 2 evaluates the performance of WAN links connecting Cape Town, Singapore, and Buenos to the Central Router/NOC, in comparison to LAN connectivity inside each branch. The objective is to examine variations in throughput, overhead, and link use.

Figure 4.9 below shows ***scenario 2***



Link Properties Used in this scenario

Link Type with id	Source	Destination	Changes made	Justification
Link1(Wan link)	Capetown Router	Central Network	Uplink/Propagation Delay 30 (ms)	WAN connections require longer delays because the distance is long.. 30 ms is set reflecting realistic latency
Link3(Wan link)	Noc	Central Network	Uplink/Downlink Propagation delay (30ms)	None
Link39(WAN)	Bueno Router	Central Network	Uplink/Propagation Delay 30 (ms)	A typical WAN latency consistent on all remote branches
LAN Links Singapore LAN (27) Buenos LAN(31) NOC LAN(21) CapeTown LAN (13)	Switch	Wired nodes within Their branch	Default	LAN connection usually cover short distance with low delay or no delay

We will simulate for 20s based on these configurations

Link Metrics

Link	Bytes transmitted(B)	Payload transmitted(B)	Overhead transmitted(B)
1	4420390	4017906	402484

Figure 4.10 below shows the result as per the link throughput vs time of ***link1***

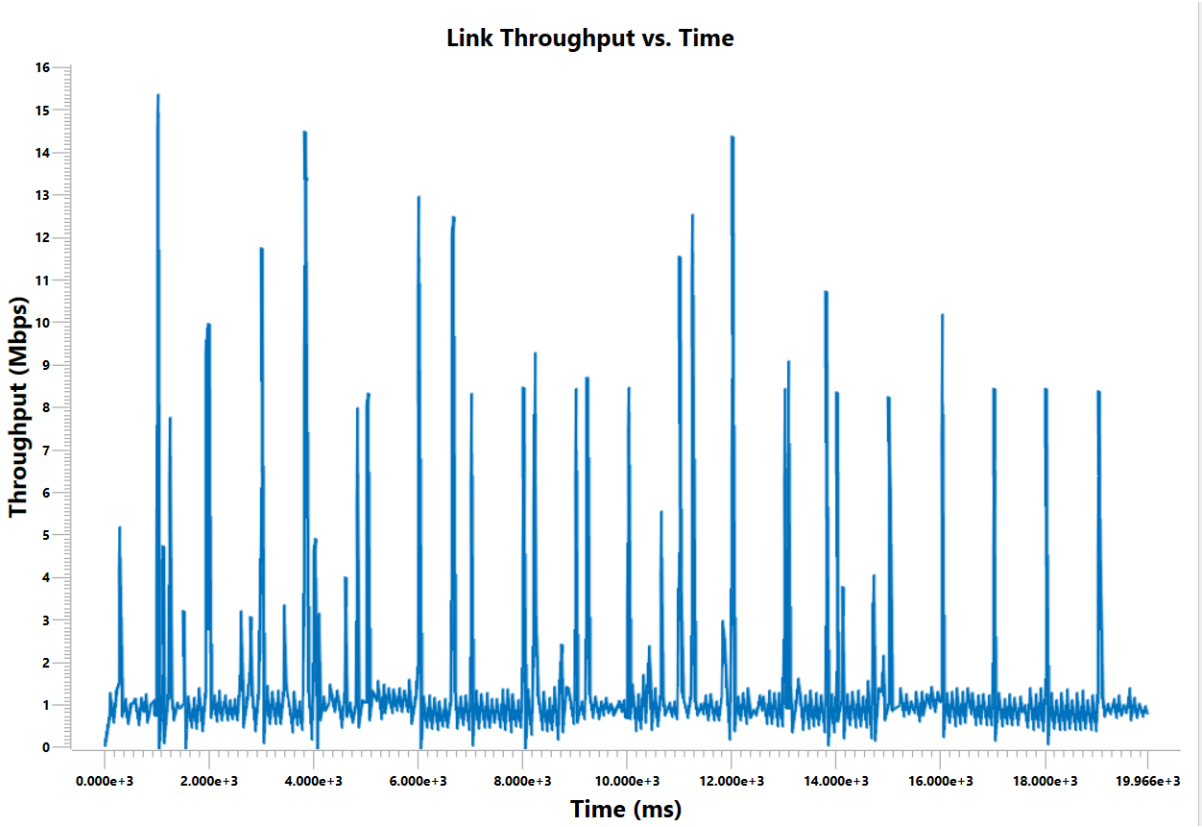


*Figure 4.11 below shows the result as per link throughput vs time of **link3***

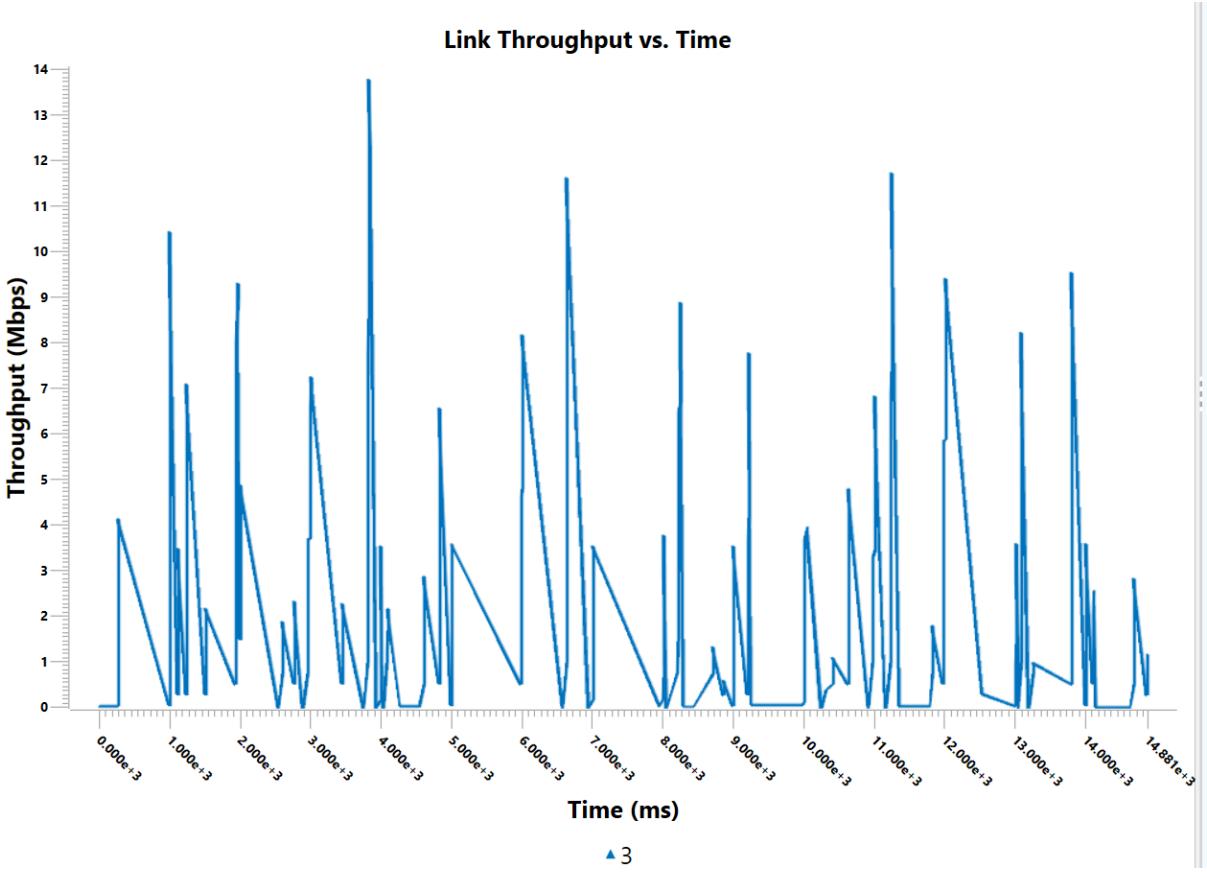


*Figure 4.11 below shows the result as per link throughput vs time **link39***

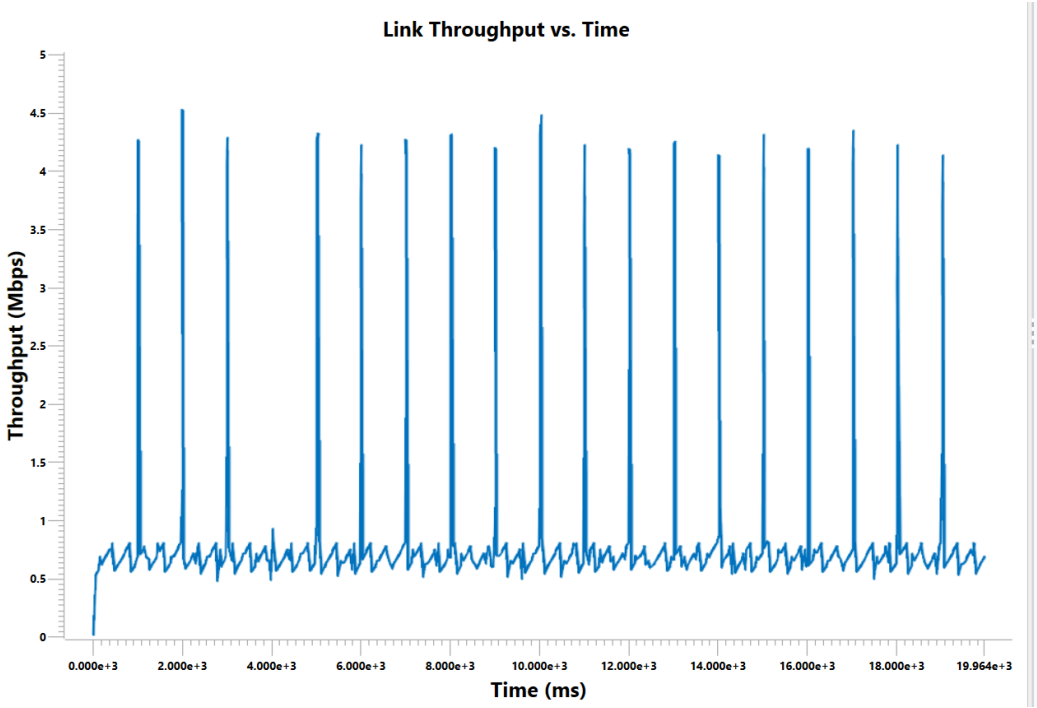
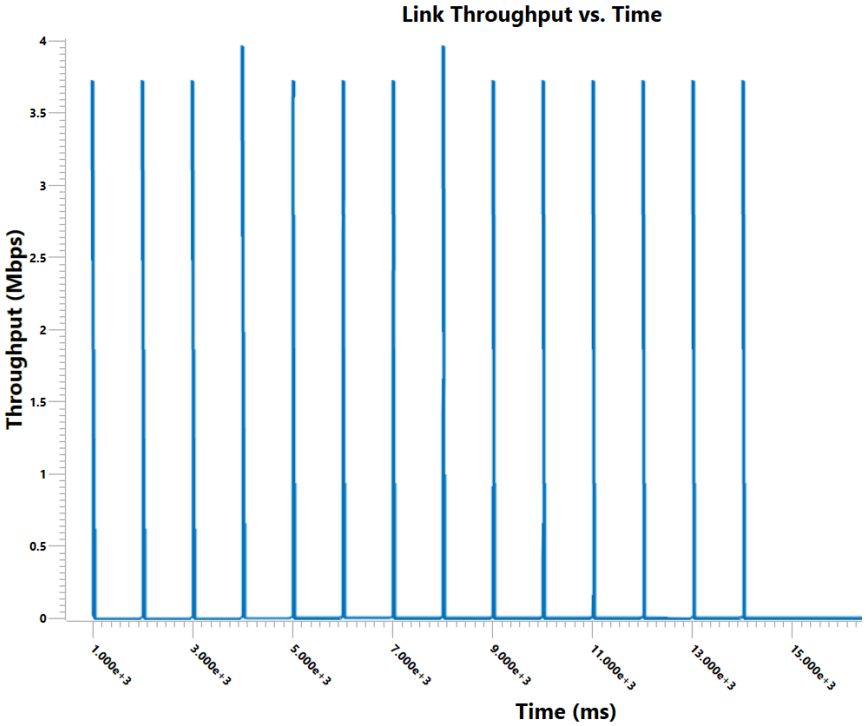


Figure 4.12 below shows the result as per the link throughput vs time ***Link31***



Link	Bytes Tx	Payload	Overhead	Notes
Singapore LAN (27)	438,576	400,000	38,576	No errors
Buenos LAN (31)	445,106	395,710	49,396	1 error
NOC LAN (21)	1,702,777	1,543,195	159,582	Highest LAN load
Cape Town LAN (13)	438,576	400,000	38,576	No errors

The results indicate that the WAN links exhibit spikes in the graph, signifying latency, overhead, and occasional errors. While LAN links exhibit smooth and consistent throughput, they render the LAN more effective than the WAN.

TASK5

i) Utilisation of the backbone link (Wan Linkid_1)

Figure 5.1 below shows a scenario

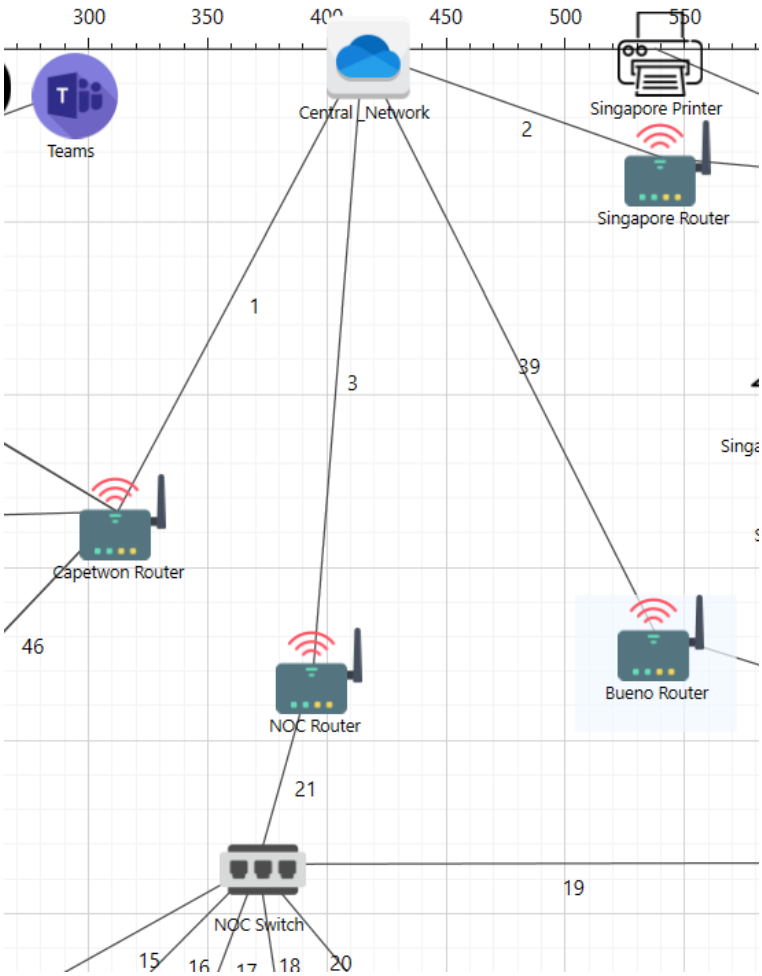
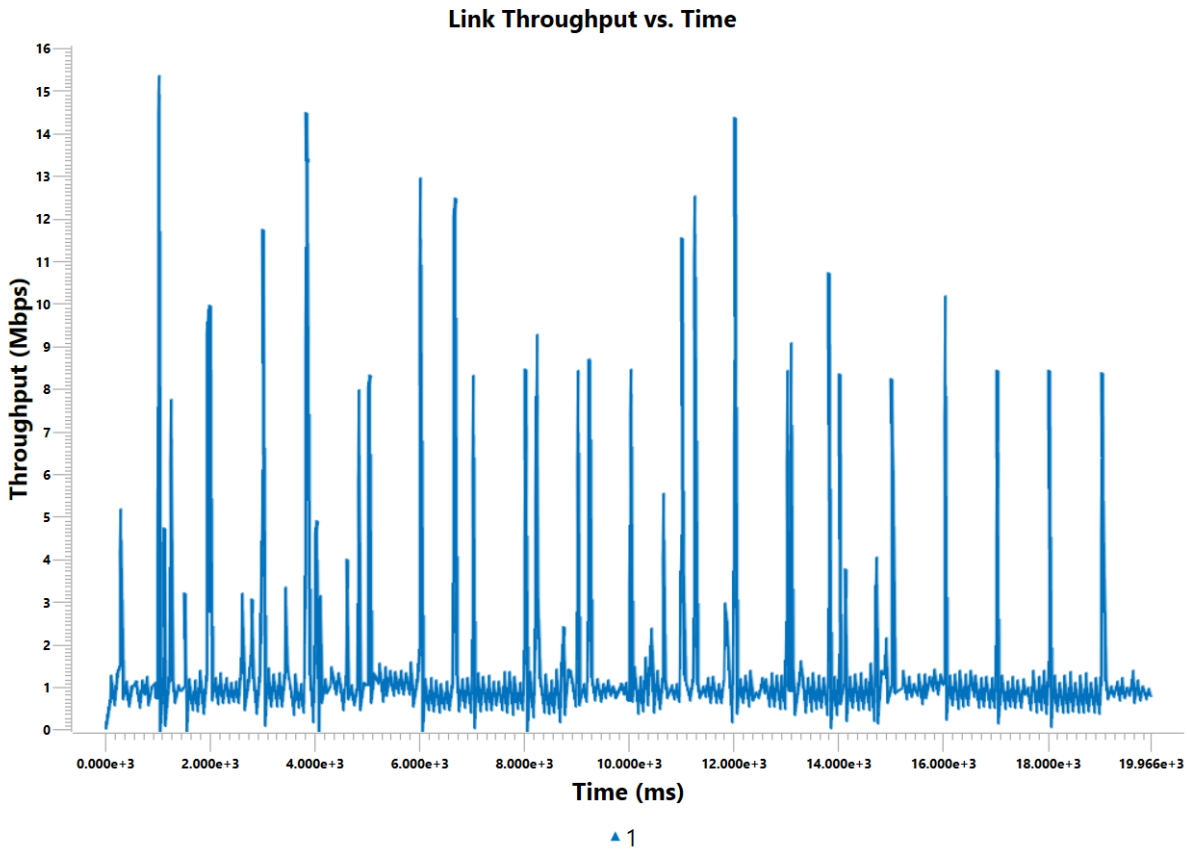


Figure 5.2 below shows the result as per link throughput vs time ***Linkid_1***

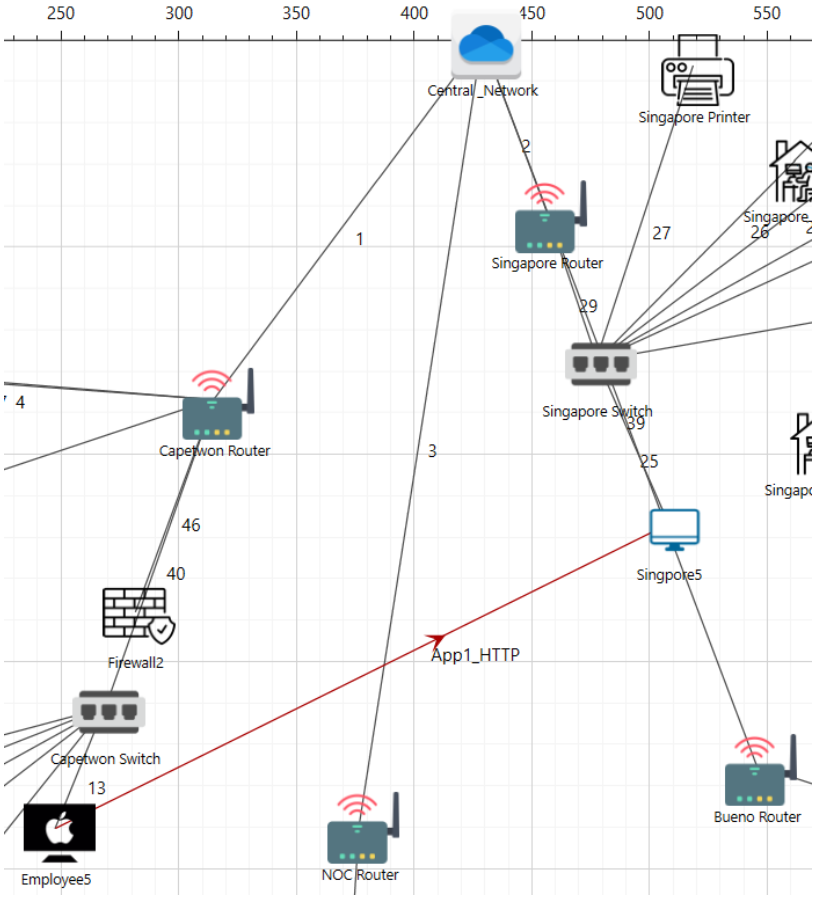


Link	Bytes transmitted(B)	Payload transmitted(B)	Overhead transmitted(B)
1	4420390	4017906	402484

Component	Role	Configurations made	Justification	Results as per Configurations
WAN link(CapeTown to Central Router)	Backbone link connecting to the Central Network	Propgation delay 30ms	This WAN link is the main backbone link, marked by the highest traffic flow owing to its hierarchical network structure. .	High utilization peaks of 0-15 Mbps indicating efficient WAN performance and highlight protocol overheads.

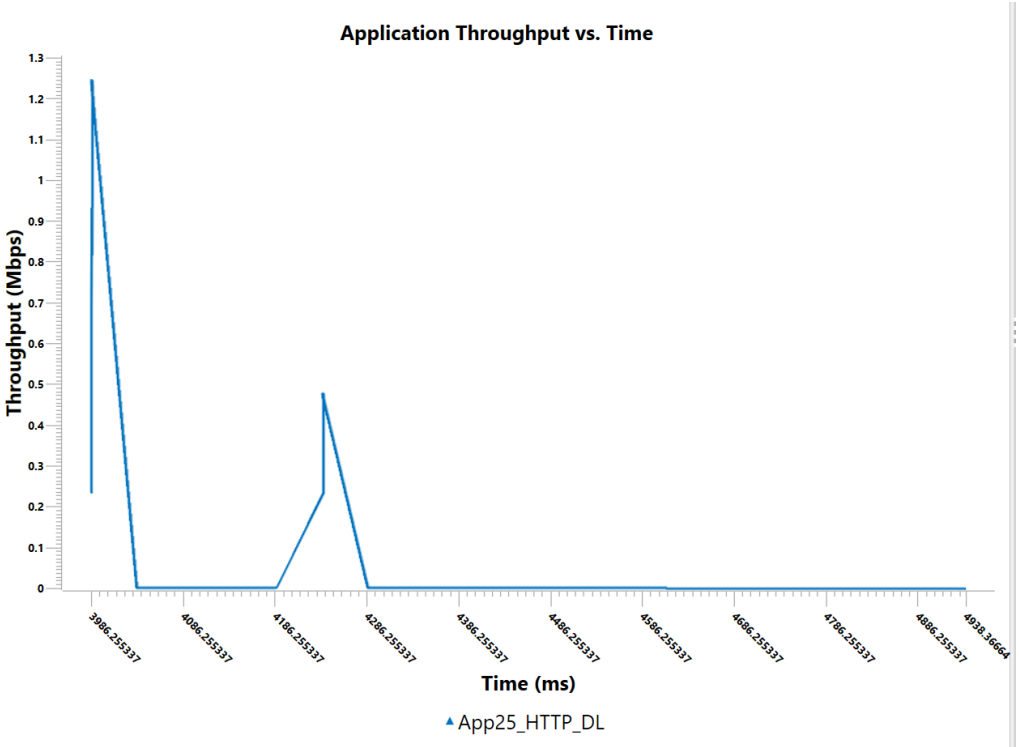
ii) OneDrive average file download time

Figure 5.3 below shows the traffic scenario



In this case, we will examine the HTTP traffic transmitted from CapeTown to Singapore, assuming the user downloads a OneDrive file shared by the CapeTown user.

Figure 5.4 below shows the result as per link throughput vs time ***App25 HTTP DL***



Link	Application	Packtes generated	Packets Received	Throughput	Delay	Jitter Microsecond	Payload generated	Payload received
25	App25_HTTP	18	18	0.0054	56252.86	117627.2	13500	13500
25	App25_HTTP	252	252	0.144	8239.414	2355.575	360000	360000

Component	Configurations made	Justification
Http Application being sent from CapeTown to Singapore	Exponential Distribution Page Inter arrival time:3s size:5000B	The traffic is modeled using an exponential inter-arrival time of 3 seconds, with each page set to 5000 bytes to simulate an

		actual document.

Download Time (s)= Throughput (Bytes/s) File Size (Bytes)

Time1= 13500/56252.86 =0.240s
Time2=360000/8239414=43.70s

Average file download time: Time1+Time2/2= 21.97s

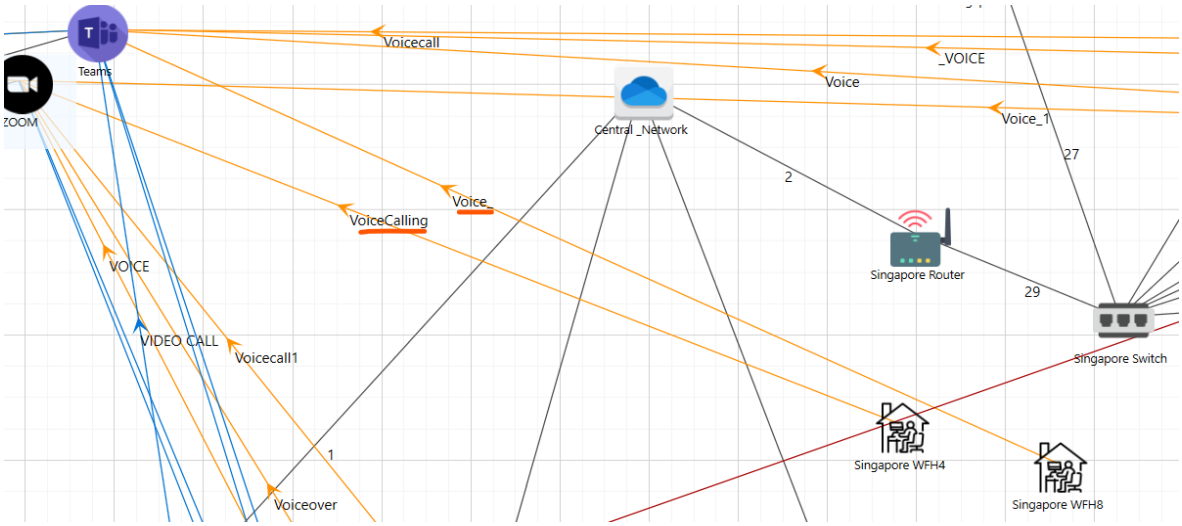
iii) The server page response time is 0 due to the immediate reaction of the cloud server. The WAN transmission delay contributes significantly to the extended overall download time. Based on **ii)average file download time**

Server response time Total download time – network transmission time

iv)

We will compare voice applications made in Teams vs made in Zoom

Figure 5.4 below shows traffic Sceanrio



In this scenario, one user is on a Microsoft Teams voice call while another is on a Zoom voice call, both participating from the Singapore branch.

Configurations

Voice_ (MS Teams)

*Figure 5.5 below shows the configuration made in **Voice_** application*

Transport Protocol	UDP
QoS	RTPS
Priority	Medium
Codec	G.711
Mean Generation Rate	64 kbps
Packet Size	
Packet Distribution	Constant
Packet Size (B)	160
Inter Arrival Time	
Service	
Type	CBR

VoiceCalling (Zoom)

*Figure 5.6 below shows the configuration made in **VoiceCalling** application*

QoS	RTPS
Priority	Medium
Codec	G.711
Mean Generation Rate	64 kbps
Packet Size	
Packet Distribution	Constant
Packet Size (B)	160
Inter Arrival Time	
Inter Arrival Time Distribution	Constant
Inter Arrival Time (µs)	20000
Service	
Type	VBR

*Figure 5.7 below shows the application throughput vs time in **Voicecalling** application*

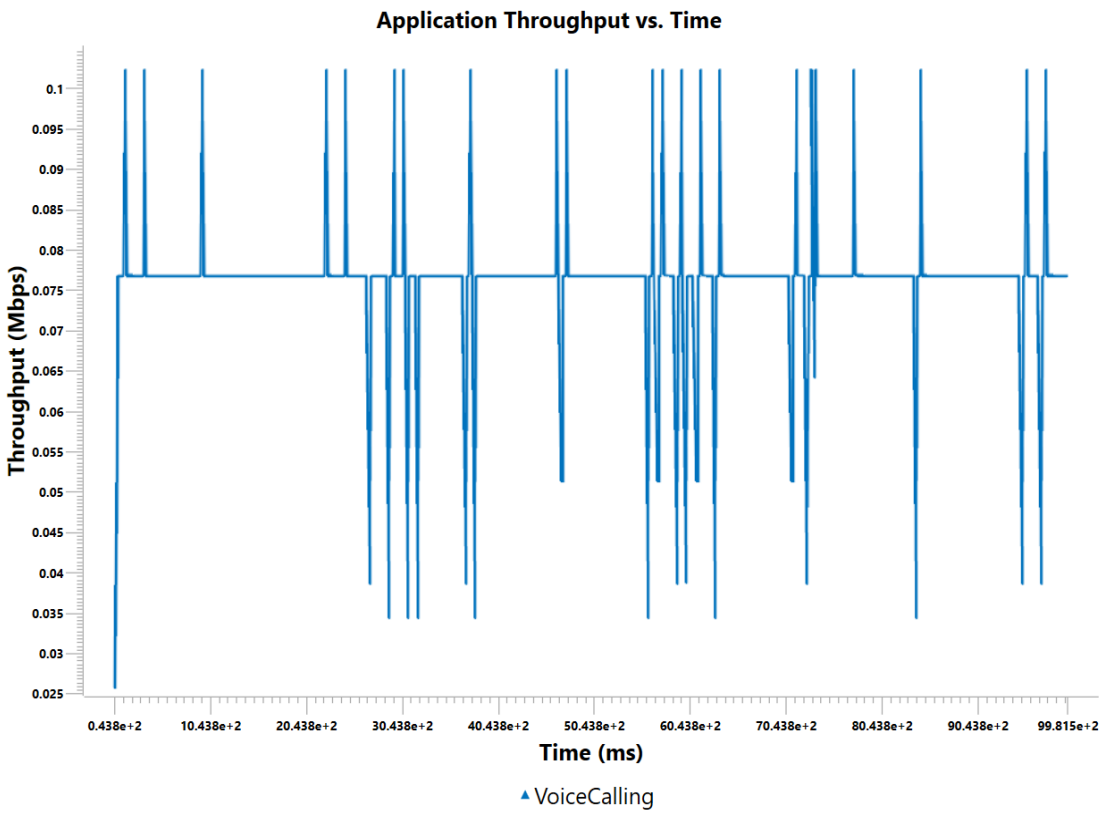
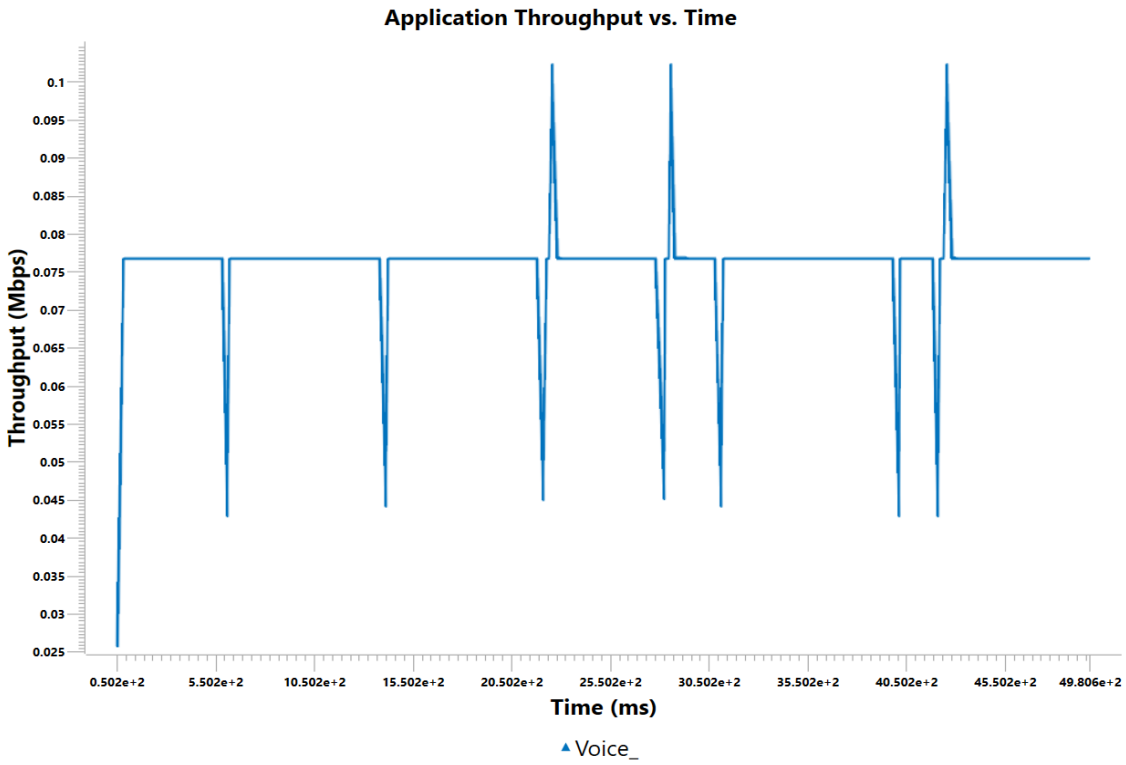


*Figure 5.8 below shows the application throughput vs time **Voice_** application*

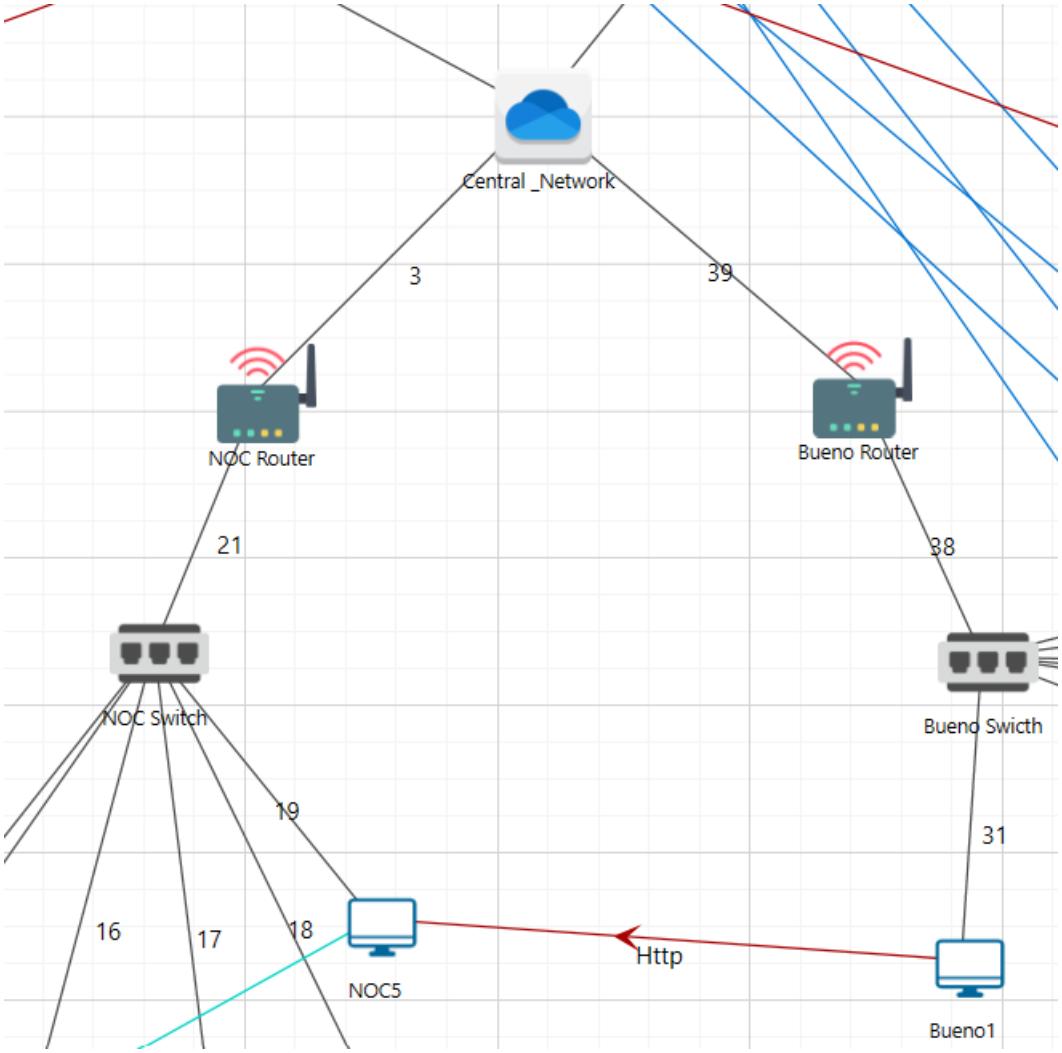


cation	Packets Generated	Packets Received	Payload generated (bytes)	Payload received (bytes)	Throughput (Mbps)	Delay (microsecond)	Jitter (microsecond)
Calling	250	248	40000	39360	0.064976	4795.632	3835.462
_	250	248	40000	39680	0.06348845	4594.833	3669.363

VBR (Zoom) introduces variability in delay and jitter, resulting in a realistic scenario whereas CBR (MS Teams) provides smoother, more predictable voice performance.

v) Bottleneck

Figure 5.9 below bottleneck sceanrio



Considering the following scenario we will demonstrate a bottleneck scenario

Configuration for analysis

Link Type with id	Changes made
WAN Links 3,39	Uplink/Propagation Delay 100 (ms)
LAN Links 21,38	Uplink/Downlink Propagation delay (30ms)
LAN links 19,31	default

Results

Figure 5.10 below shows the link throughput vs time ***in Link3***

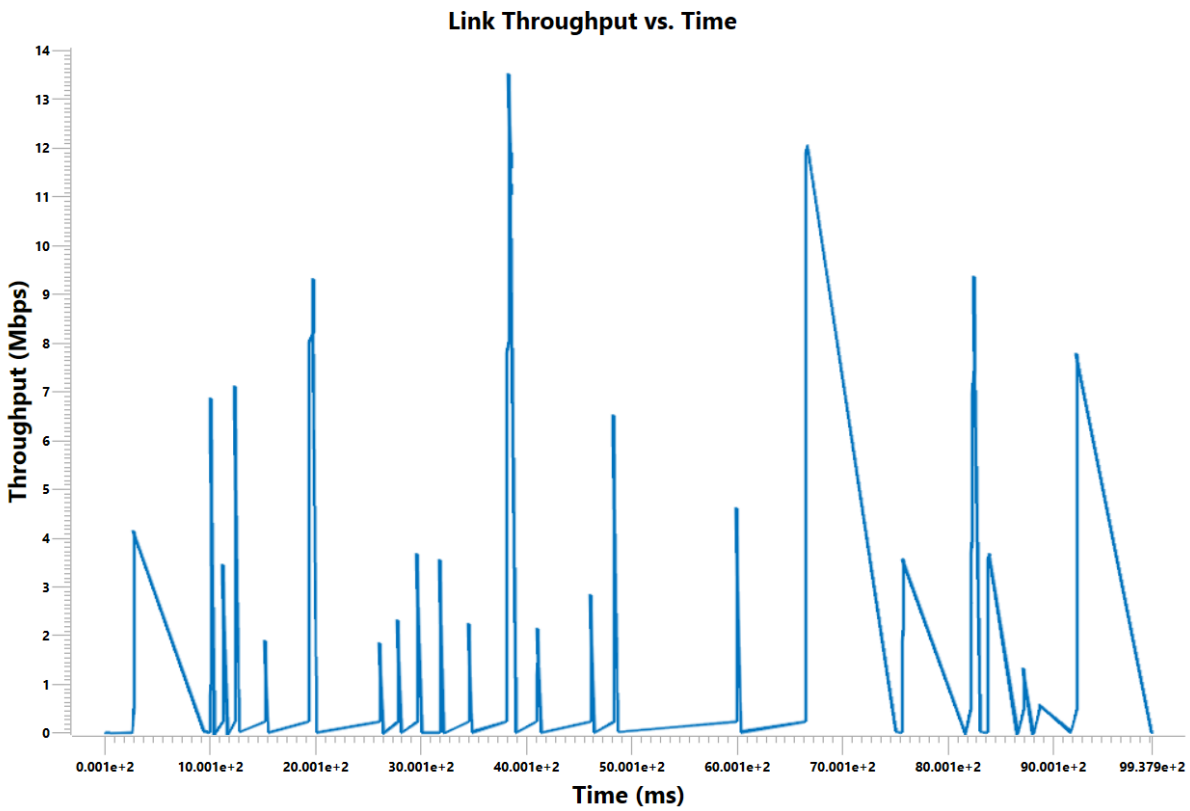


Figure 5.11 below shows the link throughput vs time ***in Link39***

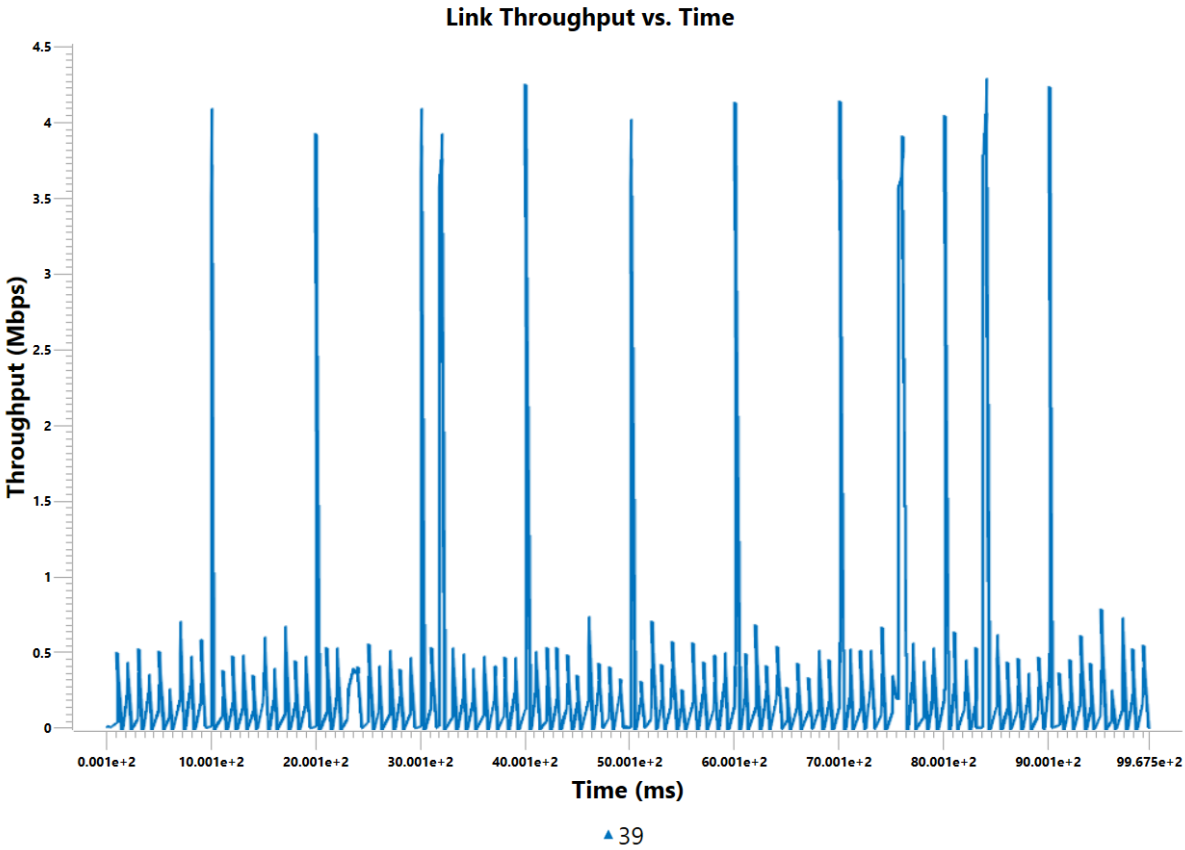


Figure 5.12below shows the link throughput vs time ***in Link38***

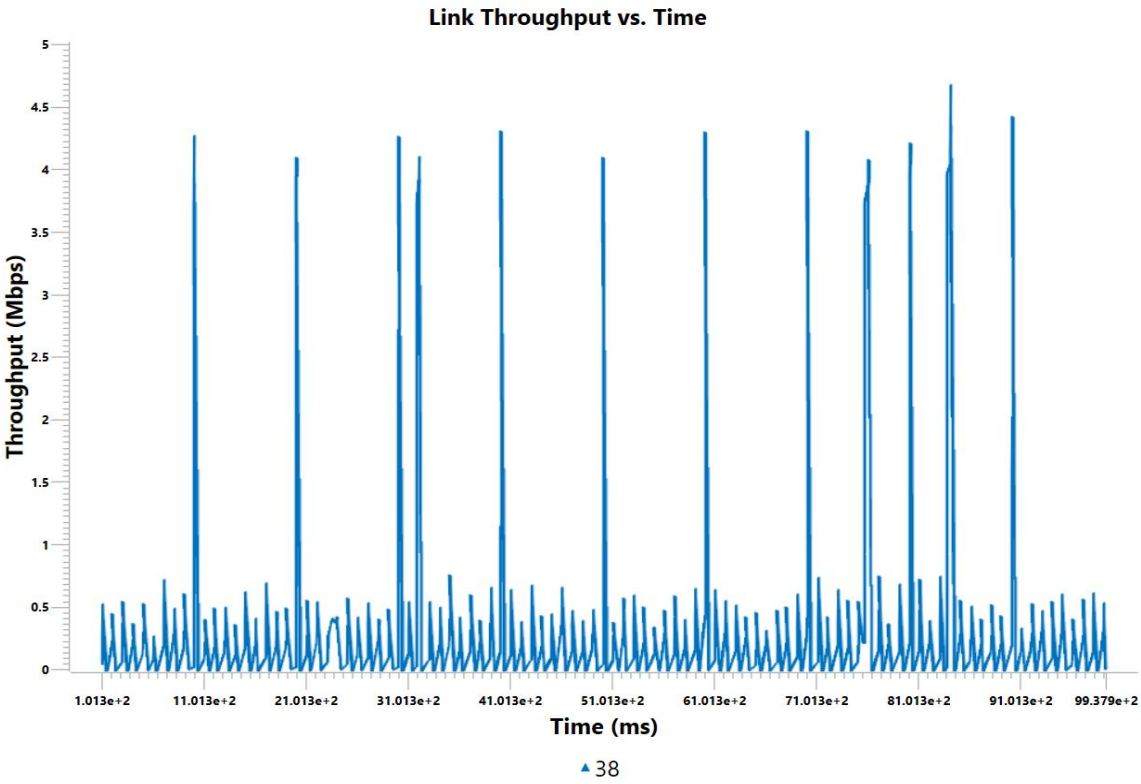


Figure 5.13 below shows the link throughput vs time ***in Link31***

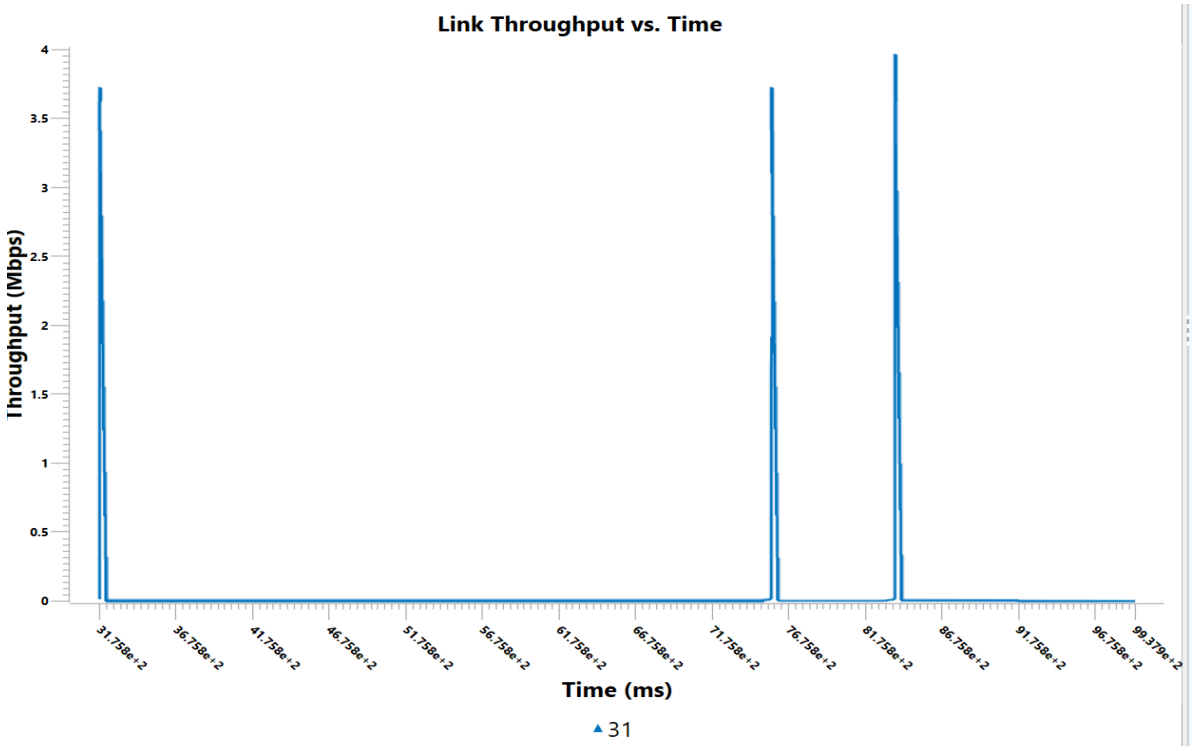
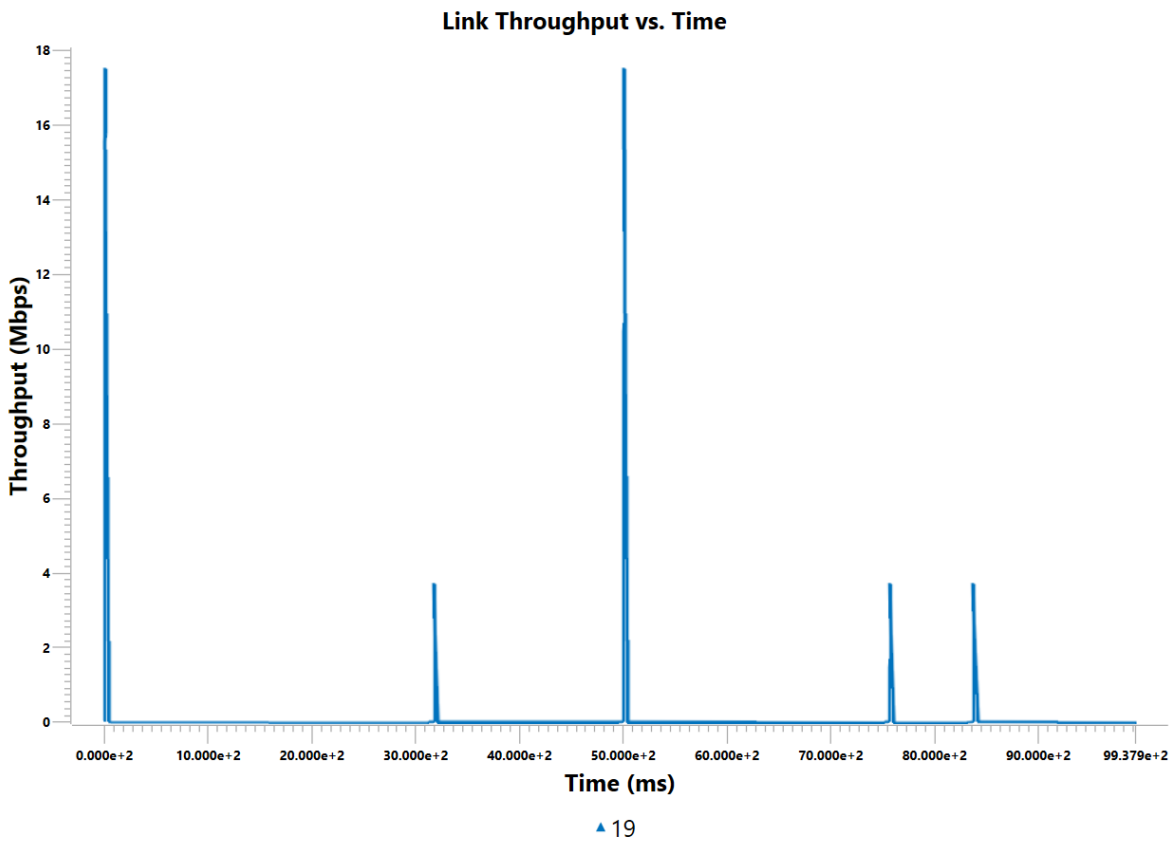


Figure 5.14 below shows the link throughput vs time ***in Link19***



Link	Packets Generated	Packets Received	Packets errored	Bits Transmitted	Payload transmitted	Overhead transmitted

					(bytes)	
39	675	278	1	631062	572921	58141
38	676	276	1	605257	571461	33796
19	183	223	0	289086	262250	26836
31	46	69	0	71324	63710	7614

Links 38 and 39 exhibit minor spikes accompanied by low throughput, resulting in packet congestion and failures, which contribute to delays and potential congestion that diminish throughput.

TASK 6

In this scenario we will illustrate comparison of meeting hosted in ms teams and zoom for this we will add 4 wired nodes in Bueno and 3 nodes in Singapore assuming they are accesing to voice/videocall

Application	users	Traffic Flow	Changes	Results
Voice call (MS Teams)	3	NOC ~ Central Network~ Singapore	Servie:CBR Distribution: Constant	CBR performance in MS teams results in smoother behavior characterized by low latency, minimal packet loss, and consistent throughput.
Voice Call (Zoom)	4	NOC~ Central Network~ Buenos Argentina	Service:VBR Distrribution Exponential	Bursty traffic leads to increased latency and jitter, resulting in slightly reduced throughput, which reflects inherent consequences.

Figure 6.1 below shows the application throughput vs time ***in _Voice (MS Teams representation)***

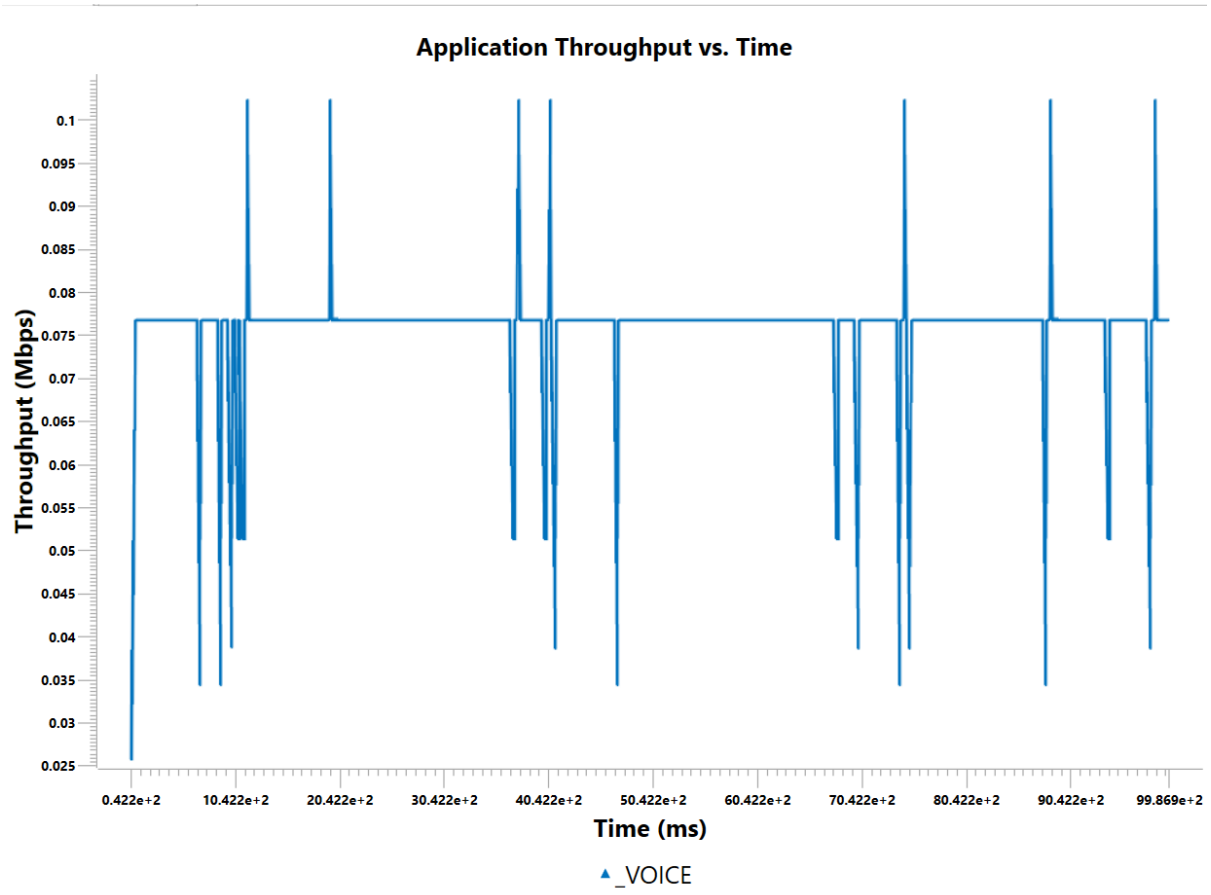
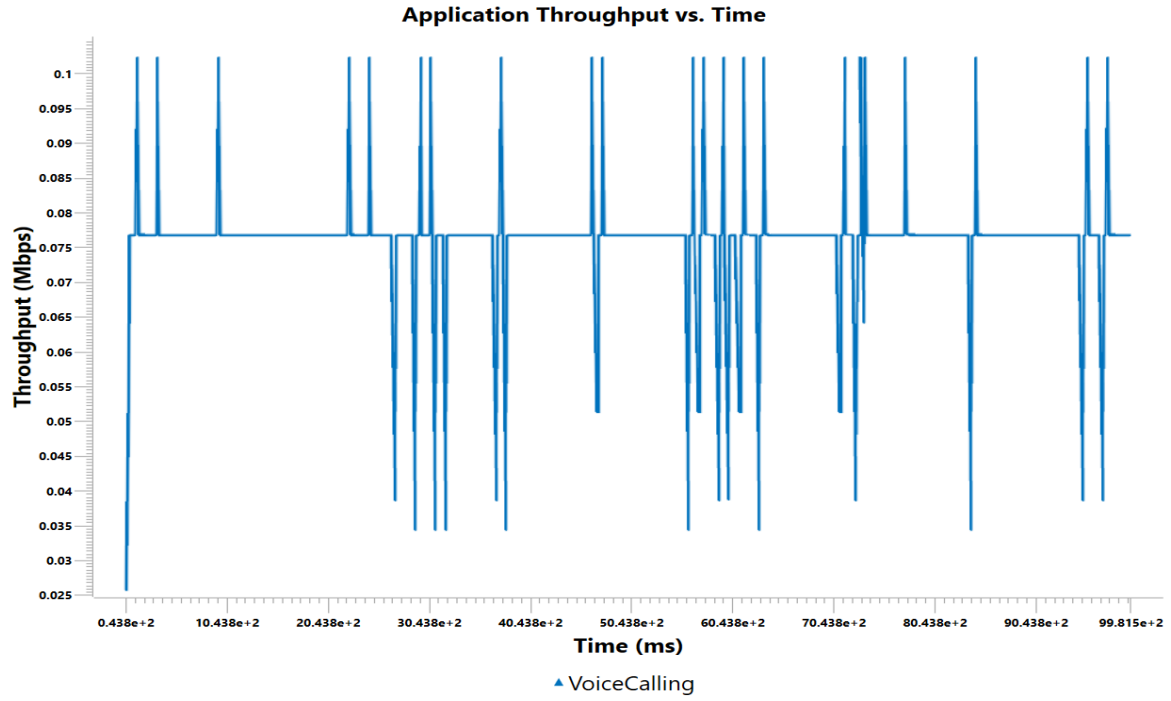


Figure 6.2 below shows the application throughput vs time ***in Voicecalling(Zoom representation)***



Application Name	Packets Generated	Packets Received	Payload generated (bytes)	Payload received (bytes)	Throughput (Mbps)	Delay (microsecond)	Jitter (microsecond)
VoiceCalling(Zoom)	250	248	40000	39360	0.064976	4795.632	3835.462
_VOICE(MSTEAMS)	250	250	40000	39680	0.06348845	4594.833	3669.363

Task 7

In task 5, significant congestion impacted the backbone link due to a 100ms propagation delay between uptime and downtime, leading to reduced throughput, packet errors, and poor application performance. Enhancements will be evaluated, and a re-execution will occur with a 30ms propagation delay to resolve these issues.

*Figure 7.1 below configurations made in **in Link39***

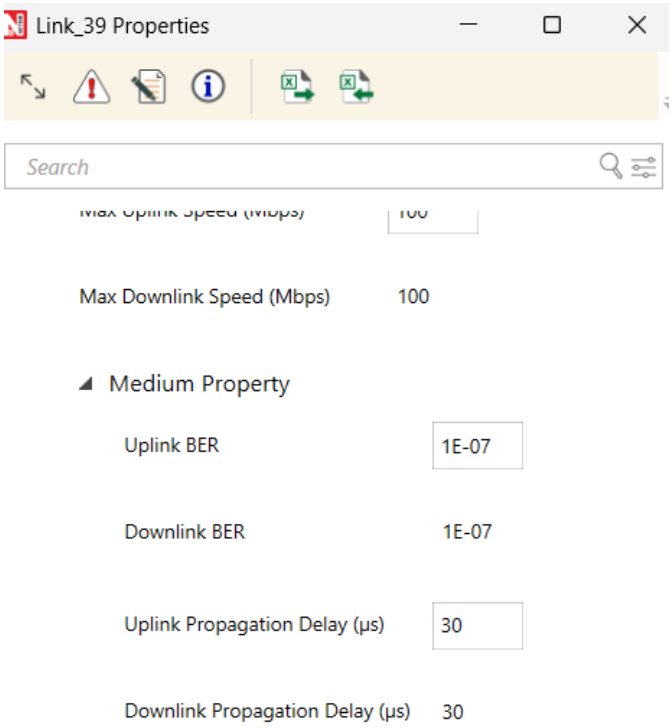
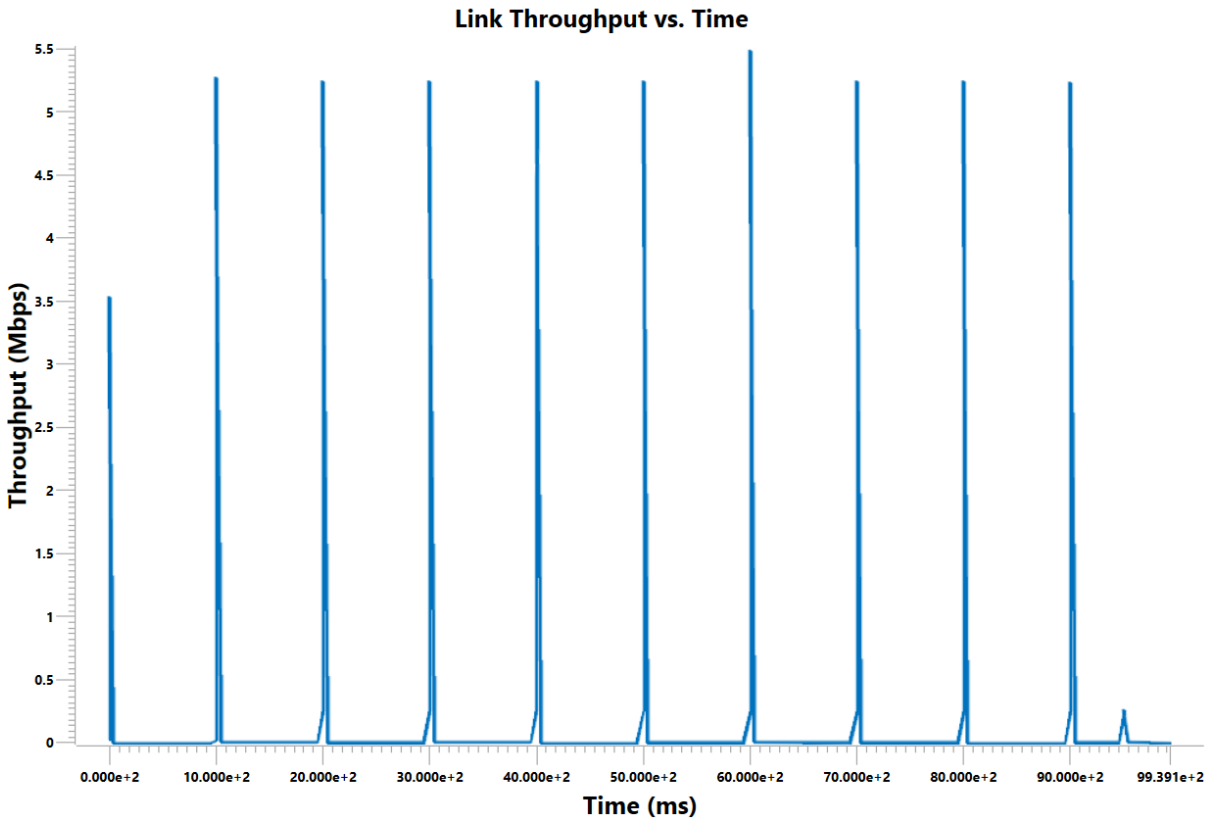


Figure 7.2 below shows the link throughput vs time ***in Link39***



Link	Packets Generated	Packets Received	Packets errored	Bits Transmitted	Payload transmitted (bytes)	Overhead transmitted
39	205	209	0	320260	292920	27340

Analysis

- Zero packet errors
- Throughput increased
- Reduced overhead
- Link stability increased

Task 8

• Conducted thorough network performance analysis involving design, application configuration, LAN/WAN behaviour analysis, and change implementation.
• Examined differences between CBR and VBR, LAN and WAN traffic, and inter-branch email/application performance.
• Identified performance bottlenecks caused by unnecessary overhead and increased transmission delays through practical simulations.
• Made modifications to link settings, streamlined routing, and optimised application traffic to reduce overhead and ensure zero packet errors.
• Achieved steady throughput after implementing changes, strengthening the link between theoretical concepts and real-world network behaviour.

Conclusion

- Technology choices, network design, and application configurations impact a global company network's performance, risk profile, and operational efficiency.
- Addressing vulnerabilities like misconfigured cloud services requires early DevSecOps integration, continuous testing, and proper configuration management.
- Small security vulnerabilities can lead to major exploitation if not managed through secure coding, static/dynamic testing, and regular patching; examples include the Jenkins Scriptler plugin vulnerability and CWE-302.
- LAN environments outperform WAN links due to shorter propagation delays, lower overhead, and reduced packet loss, despite WAN having higher throughput.
- WAN networks exhibit more volatility with longer round-trip times and protocol overheads, affecting applications differently.
- Real-time applications (e.g., speech and video) are sensitive to jitter and delay, while data applications (e.g., email, HTTP, FTP) are influenced by throughput, dependability, and overhead. The OneDrive HTTP download experiment highlights WAN latency as a critical bottleneck affecting user experience, even with low server response times.

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